Spatial explanations of OD demand effects in traffic networks: Topological conditions and exact boundary summaries

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ABSTRACT

Origin–destination (OD) demand changes affect traffic costs at many locations, but the spatial aggregation of their attributions lacks a topology-aware criterion for deciding when summaries taken at different network boundaries are comparable. We develop a topology-based theory of signed boundary accounts for differentiable traffic-assignment models and conservative learned surrogates. The main result gives necessary and sufficient conditions for cut-consistent explanations under conservative demand responses. For two-sided connected regions, the condition depends only on the block atoms traversed by the queried OD paths, allowing arbitrary off-path heterogeneity; multiple queries merge these atoms transitively. When exact cut consistency does not hold, a node-divergence representation yields closed-form account extrema, the minimax common account and its exact error, and the maximum disagreement between learned and reference explanations. The same representation supports reusable scans of prescribed screenlines and compact piecewise summaries with explicit error control. Applications to Sioux Falls and Anaheim show how free-flow and congestion weights combine with study and background rerouting to shape spatial response profiles. In a conservative learned-assignment experiment on 96 test demand paths, response supervision reduces median attribution-column and geometric-boundary errors from 21.18% and 8.74% to 14.19% and 5.75%. The framework provides mathematically controlled spatial explanations while retaining the original traffic outcome.

1. Introduction

A demand change may alter traffic costs at several locations along an origin–destination corridor, and the net crossing-cost response generally differs from one boundary position to the next. The object called for is therefore not a single network-wide number but a spatial response profile: how the signed crossing-link cost contribution evolves across screenlines and regional boundaries, and which traffic adjustments explain that evolution. We retain the original edge-level attribution and develop a framework that compares boundary accounts, determines when they can share a common summary, and resolves their differences into spatial and traffic-response components.

Transportation sensitivity analysis asks how changes in origin–destination (OD) demand affect link flows and system outcomes (Tobin and Friesz, 1988; Josefsson and Patriksson, 2007; Patriksson and Rockafellar, 2003). For a differentiable assignment model or learned surrogate, integrated gradients allocate a finite outcome change across OD demands (Sundararajan et al., 2017). Resolving this allocation through individual link flows additionally locates the contribution on the network — and raises the spatial question above, because different aggregations of the same edge attributions need not agree.

Consider a planner tracking one demand change across several predetermined corridor cross-sections or regional boundaries, using a fixed road partition. The task is to describe how the signed crossing-cost response varies across these positions, decide which positions can share one concise summary, and explain which links and traffic responses cause the summaries to differ. The inputs are the fixed network, demand change, outcome function and candidate boundaries; the outputs are their OD accounts, a transferability criterion, and an additive explanation of their differences.

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The network-wide contribution C_j, attribution on roads within a region, and signed boundary account $\phi_j(U)$ answer distinct questions. Different boundaries define different outcome functions, so equality is a structural property to establish rather than a prerequisite for attribution. The proposed classification determines when the summaries share a common boundary value across conservative responses. When they do not, a node representation computes their differences and identifies the traffic responses responsible.

The question differs from tracking physical OD vehicles through screenlines or cordons (Sheffi, 1985; van Zuylen and Willumsen, 1980; Bar-Gera, 2002): an OD-indexed sensitivity describes the network-wide response, including other ODs' route changes. Flow conservation transfers to flow-valued attribution, but outcome weighting introduces additional topology-dependent conditions (Section 3).

This distinction matters for explanations of conservative traffic models: physical penalties, output projections, and exactly conservative parameterizations constrain the predicted flows (Raissi et al., 2019; Shi et al., 2021; Valente et al., 2025; Richter-Powell et al., 2022), but conservation of those flows does not by itself make a learned model's attribution transferable across boundaries. Exact conservation transfers to flow-valued integrated gradients; outcome weighting then imposes additional conditions on the network and the outcome definition; and approximate output conservation may fail at the first transfer step, because a small residual need not have small OD derivatives.

This paper makes three contributions. First, we give a query- and region-dependent topological characterization of cut-consistent OD attributions. For two-sided connected regions, the required weight equalities are confined to the block atoms traversed by the queried OD paths, while multiple queries merge these atoms transitively. Second, we develop an exact node-divergence representation that yields boundary-account differences, unrestricted extrema, the minimax common account and its error, prescribed-screenline scans, and exact learned-versus-reference disagreement. Third, we use the framework to explain spatial response profiles in Sioux Falls and Anaheim and to evaluate conservative learned assignment models. The experiments identify how study traffic, background rerouting, free-flow costs and congestion jointly shape the profiles, and show that response supervision improves spatial explanations beyond flow prediction alone. The novelty lies in coupling block structure with the queried OD directions, the admissible region family, and the fixed outcome-weighted attribution object.

2. Related work

Attribution and cost allocation. Our edge-level object is input-specific conductance (Dhamdhere et al., 2019; Lundstrom et al., 2022): demand is the input, link flow is an intermediate variable, and the traffic outcome is the output. Its sum over links gives the integrated-gradient contribution of an OD. Continuous marginal cost allocation also underlies Aumann–Shapley sharing (Aumann and Shapley, 1974; Billera et al., 1978). In transportation applications, attribution typically explains a prediction model by ranking its input features (Sarıyer et al., 2024; Guo et al., 2024); the object attributed here is instead a link flow that must satisfy node balance. Those scalar allocations and computational-graph completeness do not impose that balance on a traffic network. We analyze this additional property.

Conserved attribution flows on graphs. Graph-based explanation methods construct attribution flows with boundary-consistency or conservation properties on directed acyclic graphs: Shapley Flow makes edge attributions consistent across different choices of explanatory boundary, with conservation at internal nodes following from the efficiency axiom together with boundary consistency, and the recursive Shapley value recursively constructs edge-attribution flows that conserve at internal nodes (Wang et al., 2021; Singal et al., 2021). Those works design new attribution rules satisfying such axioms on computational or causal graphs. Here the graph is a physical traffic network with cycles, the edge-level conductance is fixed by the assignment model and outcome, and the question is when these existing outcome-weighted sensitivities have equal signed sums across OD-separating cuts, uniformly over conservative responses, rather than when a newly constructed rule is conserved.

Traffic models and sensitivity. Equilibrium sensitivity analysis studies demand-to-flow derivatives and their aggregation (Tobin and Friesz, 1988; Yang and Bell, 2007; Josefsson and Patriksson, 2007; Patriksson and Rockafellar, 2003). Learned assignment maps, implicit optimization layers, and logit equilibrium

metamodels make such derivatives available to explanations (Liu et al., 2023; Amos and Kolter, 2017; Lassen et al., 2025). Fully differentiable formulations of assignment and of network simulation now propagate these gradients end to end (Li and Nie, 2026; Seo, 2026), and heterogeneous graph-attention surrogates extend learned assignment to multi-class vehicles while embedding node-based flow conservation directly in the training loss (Liu and Meidani, 2024, 2025). Physical OD tracking through assignment and cordon counts concerns commodity flows (Sheffi, 1985; van Zuylen and Willumsen, 1980; Bar-Gera, 2002); it does not by itself establish transferable budgets for marginal-cost-weighted attributions.

Learned and differentiable traffic assignment. Existing studies primarily evaluate flow accuracy, computational efficiency and conservation of predicted flows (Liu et al., 2023; Liu and Meidani, 2024, 2025; Li and Nie, 2026). The present work addresses a different question: when an OD-indexed outcome attribution can be transferred across physical network boundaries, and how spatial explanation error can be quantified when that transfer is not exact.

Conservation and graph structure. Physics-informed losses and conservative architectures constrain physical outputs (Raissi et al., 2019; Shi et al., 2021; Richter-Powell et al., 2022); projections can enforce constraints exactly (Valente et al., 2025; Baez et al., 2024). We examine their transfer to flow and outcome attributions; for approximate conservation, derivative bounds of Landau–Kolmogorov type are classical (Shadrin, 2014), and Appendix A.4 transfers such control to attribution residuals. The relevant circulation space and its decomposition into circuit classes are classical (Oxley, 2011; Hopcroft and Tarjan, 1973). The new classification couples this decomposition to queried OD directions and their transitive equality constraints.

3. Attribution objects and balance criteria

3.1. Edge attribution and the accounting question

Let $\mathcal{G} = (V, E)$ be connected, with incidence matrix $B \in \{-1, 0, 1\}^{n\times m}$ (negative at a link's tail), OD vectors $\mu_j = e_{t_j} - e_{s_j}$, $s_j \neq t_j$, and $M = [\mu_1, \dots, \mu_K]$. Reciprocal and parallel arcs remain distinct edges when orientations are forgotten. Let $f : \Omega \subset \mathbb{R}^K \to \mathbb{R}^m$ map demands to flows on an open demand domain Ω, and $T : \mathbb{R}^m \to \mathbb{R}$ be a specified edgewise outcome. Assume both maps are C^1 on open neighborhoods of the relevant paths. With $\Delta q = q^1 - q^0$ and $\gamma(\lambda) = q^0 + \lambda\Delta q$, define

$$G_{:,j} = \Delta q_j \int_0^1 J_f(\gamma(\lambda))\mathbf{e}_j \, d\lambda, \tag{1}$$

$$\widetilde{C}_{:,j} = \Delta q_j \int_0^1 D_\lambda J_f(\gamma(\lambda))\mathbf{e}_j \, d\lambda, \quad D_\lambda = \operatorname{diag}(\nabla T(f(\gamma(\lambda)))), \tag{2}$$

$$C_j = \mathbf{1}^\top \widetilde{C}_{:,j}, \qquad \sum_j C_j = T(f(q^1)) - T(f(q^0)). \tag{3}$$

G attributes flow changes; $\widetilde{C}$ resolves the outcome contribution through links; C_j is its scalar OD contribution. Their columns are signed system sensitivities, including other ODs' route adjustments, rather than physical commodity flows. Entry $\widetilde{C}_{e,j}$ is input-specific conductance through link e (Dhamdhere et al., 2019).

If $Bf(q) = Mq$, differentiation and integration give

$$BJ_f(q) = M, \qquad BG_{:,j} = \mu_j \Delta q_j. \tag{4}$$

The same conclusion holds for $Bf(q) = Mq + b$ with constant background demand balance b. Outcome attribution is *OD cut consistent* when $B\widetilde{C}_{:,j} = \beta_j \mu_j$ for some $\beta_j \in \mathbb{R}$. The scale β_j is a boundary value, generally different from C_j: on two series links with unit weights, $C_j = 2\Delta q_j$ but $\beta_j = \Delta q_j$. Outcome weighting is what makes cut consistency a genuine restriction: on two series links, a conservative unit response $(1, 1)$ with weights (w_1, w_2) balances at the intermediate node only if $w_1 = w_2$, although the response itself conserves exactly.

Table 1
Three spatial summaries of one OD attribution. E_R is a road set specified before aggregation.

Object	Transportation question
$C_j = \sum_e \widetilde{C}_{e,j}$	How much does this OD contribute to the system outcome change?
$\sum_{e\in E_R} \widetilde{C}_{e,j}$	How much of that contribution is located on the specified regional roads?
$\phi_j(U)$	What is its signed aggregation on the specified boundary?

For $s_j \in U \subseteq V \setminus \{t_j\}$, orient the cut account as $\phi_j(U) = -\mathbf{1}_U^\top B\widetilde{C}_{:,j}$, and write $b_j = B\widetilde{C}_{:,j}$. Any two separating sets satisfy

$$\phi_j(U') - \phi_j(U) = -\sum_{v\in U'\setminus U} b_{jv} + \sum_{v\in U\setminus U'} b_{jv}; \tag{5}$$

for nested sets $U \subseteq U'$ the second term vanishes, leaving $\phi_j(U')-\phi_j(U) = -\sum_{v\in U'\setminus U}(B\widetilde{C}_{:,j})_v$. The difference between two boundaries is thus the enclosed imbalance, with incoming and outgoing regions entering with opposite signs. For a separable outcome $T(F) = \sum_e g_e(F_e)$, the account itself also has a direct reading. With $\delta^+(U)$ and $\delta^-(U)$ the arcs leaving and entering U,

$$H_U(F) = \sum_{e\in\delta^+(U)} g_e(F_e) - \sum_{e\in\delta^-(U)} g_e(F_e), \qquad \phi_j(U) = \mathrm{IG}_j[\,H_U \circ f\,], \tag{6}$$

because each entry $\widetilde{C}_{e,j}$ of Eq. (2) is the integrated gradient of the scalar outcome $g_e \circ f$, and the incidence signs turn $-\mathbf{1}_U^\top B\widetilde{C}_{:,j}$ into the signed boundary sum of H_U. For TSTT, $\phi_j(U)$ therefore attributes the demand change to the excess of the total travel cost on the region's outgoing arcs over that on its incoming arcs: a boundary-related outcome in its own right, neither the region's internal cost nor a vehicle count.

Proposition (cut-consistency equivalence)**.** *OD cut consistency holds exactly when every separating cut has one common signed value $\phi_j(U) = \beta_j$.*

Indeed, balance implies this identity. Conversely, comparing $\{s_j\}$ with $\{s_j, v\}$ forces zero divergence at every interior node; the two endpoint entries are then opposite because incidence columns sum to zero.

The road partition is part of the accounting object: subdivision changes which crossing-link costs a boundary includes (Example A.8).

Throughout, $\mathcal{U}_j^{\mathrm{all}}$ contains all sets separating s_j from t_j; $\mathcal{U}_j^{\mathrm{conn}}$ restricts both induced sides to be connected in the underlying undirected network; and $\mathcal{U}_j^{\mathrm{geo}}$ is a specified finite family of straight screenline regions, filtered for two-sided connectivity. The unrestricted extrema below are closed-form exact values. Sioux Falls connected extrema use complete enumeration. Geometric extrema and Anaheim's selected connected-region ranges are exact only over their stated finite families, and do not determine extrema over all connected regions.

3.2. Three levels of consistency

We distinguish three questions, each with its own mathematical object and test. *Structural cut consistency* asks whether all conservative responses at fixed weights share one common account on an admissible region family; it is a property of the network, weights, queries and family, classified in Section 4. *Realized cut consistency* asks whether one computed attribution column is constant on its stated family; it is read from the node divergence or the family's account range. *Explanation disagreement* asks how far a learned column's boundary accounts are from their reference; it is measured by the boundary seminorm of Section 5 or a family scan. Table 2 summarizes the correspondence; the finer family-by-family view appears in Table 3. A realized attribution can balance without satisfying the universal structural conditions; pointwise balance along a demand path implies integrated balance, and the converse can fail through cancellation. Appendix A.3 collects these examples, the restricted-response variants, and the directionality conditions under which weaker balance notions coincide with the classified property.

Table 2
Three questions about spatial transferability and their tests.

Question	Object	Test
Transferable for all conservative responses	structural condition	topological and weight classification (Section 4)
Current attribution constant on the family	realized column	node divergence or family range (Section 5)
Learned versus reference explanation	column difference	boundary seminorm $\mathcal{K}_j$ or family scan

4. Topology, OD queries and admissible regions

Take the block-cut tree of the underlying undirected multigraph. Its non-bridge blocks, together with each bridge as a singleton, are the *cycle atoms*. They are the classical graphic-matroid circuit classes, with each bridge retained separately. The *OD block path* $\mathcal{P}(s,t)$ is the atom sequence on the unique block-cut-tree path between the OD endpoints. For several queries, merge all atoms traversed by each $\mathcal{P}_j = \mathcal{P}(s_j, t_j)$ and take the transitive closure. The resulting edge sets $\bar{E}_1, \ldots, \bar{E}_{K'}$ are the *OD-compatibility classes*; atoms on no queried path remain separate.

4.1. Main result: two-sided connected traffic regions

Traffic analysis typically aggregates on screenlines, cordons and regional boundaries whose two sides are each spatially connected. The main theorem shows that this practice buys a substantial freedom in when boundary summaries transfer. Let

$$\mathcal{U}_{st}^{\mathrm{conn}} = \{U \subset V : s \in U,\ t \notin U,\ G[U] \text{ and } G[V \setminus U] \text{ are connected}\},$$

where connectivity uses the underlying undirected multigraph. These are separating bonds: both sides of the boundary form connected regions.

Main implication. For two-sided connected traffic regions, transferable OD boundary explanations require uniform marginal weighting only along the block atoms traversed by the queried OD paths. Heterogeneity elsewhere in the network is immaterial to this guarantee.

Theorem 1 (two-sided connected-region classification)**.** *For a connected loopless multigraph, $\mu = e_t - e_s$, and $D = \mathrm{diag}(w)$, the property*

$$\exists \alpha \in \mathbb{R} \quad \forall x \text{ with } Bx = \mu \quad \forall U \in \mathcal{U}_{st}^{\mathrm{conn}} : -\mathbf{1}_U^\top BDx = \alpha \tag{7}$$

holds if and only if every edge in the atoms of $\mathcal{P}(s,t)$ has one common weight α. Weights of edges outside those atoms are arbitrary, even within an off-path cycle atom.

The proof in Appendix A.6 uses connected prefix and suffix regions within each traversed atom. Each admissible boundary crosses edges in just one such atom, which explains why off-path heterogeneity is invisible to this region family. This is a universal fixed-weight statement; agreement for one realized response need not imply its conditions.

Corollary 2 (several queries on connected regions)**.** *At one shared weight vector, Eq. (7) holds for all queried ODs if and only if edges in each group of atoms joined by the queried block paths have one common weight. An atom on no queried path has no weight restriction, including no within-atom restriction.*

Thus restricting attention to connected traffic regions permits additional outcome heterogeneity. It changes the compatibility classification, while the unrestricted extrema and error formulas of Section 5 retain their own stated boundary family.

4.2. Design implications

The theorem gives three design freedoms, each checkable before any model is fitted.

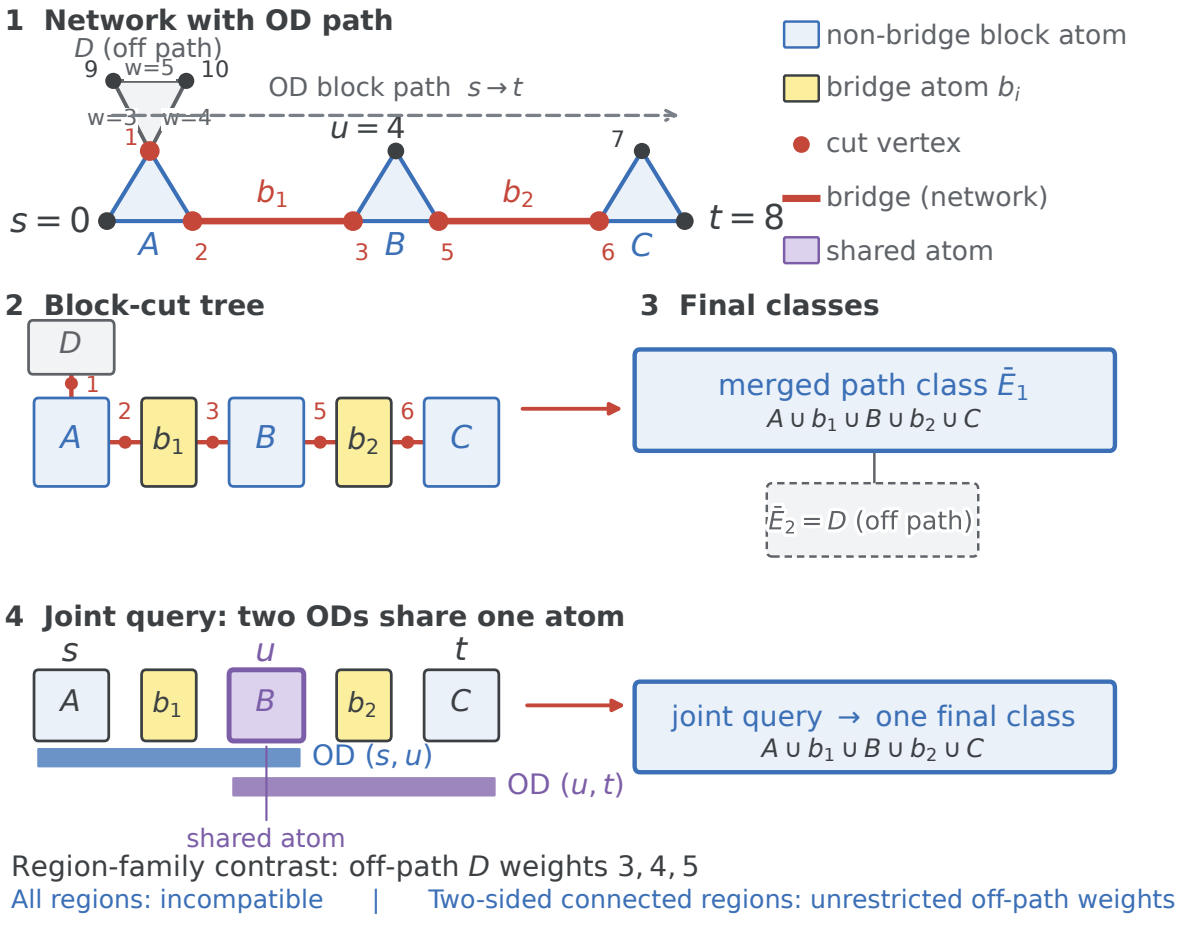


Figure 1: OD queries merge the atoms they traverse. Two paths sharing atom B merge the full chain transitively; off-path atom D stays separate. For the connected-region comparison, atom D has heterogeneous weights $3, 4, 5$: these are invisible to the queried connected boundaries, but violate unrestricted transferability.

Query design. Only the atoms traversed by the queried OD paths enter the condition, and adding queries merges atoms transitively through shared blocks. On Figure 1 ($s = 0$, $u = 4$, $t = 8$, linear outcome $T_w(F) = w^\top F$), the query (s, u) joins A, b_1, B while b_2, C and D keep their own constants; adding (u, t) joins the remaining chain through the shared atom B; adjusting only b_2 and C then restores a common account for both queries while the unrelated atom D keeps weight 3. Appendix A.2 tabulates the three stages with exact account ranges.

A constructive distinction on Figure 1. Keep weights 1 on A, b_1, B, weights 2 on b_2, C, but set the three off-path D weights to $3, 4, 5$. The first query (s, u) admits common account 1 on every two-sided connected region for every conservative response. The second query (u, t) initially has connected accounts spanning $[1, 2]$ and fails that criterion. Changing only b_2, C to weight 1 makes both queries compatible on connected regions, while D remains internally heterogeneous. For each query there are 512 unrestricted separating sets but only 5 two-sided connected sets. Checking a particular response and a complete circulation basis confirms the universal connected-region assertions; this is stronger than checking sampled responses. A unit signed circulation in D gives unrestricted accounts $[-1, 3]$ after the merge, while all connected accounts remain 1. Thus the unrestricted criterion still fails. This fixed structural outcome illustrates the theorem; its weights do not replace the original TSTT marginal costs. The reproducible record is provided with the code.

Outcome design.

Corollary 3 (linear outcome functionals)**.** *Let $f \in C^1$ be exactly conservative on an open neighborhood of the attribution path, and let $T(F) = a^\top F$ be a linear functional of realized flows. Then $\nabla T \equiv a$, so structural cut consistency over the unrestricted separating family holds if and only if a is constant on the compatibility classes, with common cut account $\beta_j = \Delta q_j\, a|_{\bigcup \mathcal{P}_j}$; over two-sided connected regions the weaker condition of Theorem 1 applies instead — a constant on the block atoms traversed by the queried OD paths, with no constraint on off-path atoms. Both are conditions on the outcome definition alone, checkable by inspection before any model is trained, and the pointwise classification holds at every λ.*

Nonlinear compatible outcomes also exist: if $T(F) = \Phi(\sum_{e\in\bar{E}_1} F_e, \dots, \sum_{e\in\bar{E}_{K'}} F_e)$, its gradient is class-constant at every flow. More familiar traffic outcomes require equality of their actual marginal weights. The Beckmann potential has weights $t_e(F_e)$; total system travel time (TSTT) has weights $t_e(F_e) + F_e t'_e(F_e)$; a separable index $\sum_e g_e(F_e)$ has weights $g'_e(F_e)$. Heterogeneous link parameters and congestion typically violate these equalities. A single-link outcome has a nonzero common boundary value only when the queried OD block path is precisely that singleton bridge.

Region-family design. The admissible family is part of the accounting question, and the theorem quantifies what each choice costs: the connected family ignores off-path heterogeneity, the unrestricted family does not, and a prescribed finite family such as a screenline set admits exact realized checks but no structural classification beyond the connected bound (Section 5, Table 3).

4.3. A stronger criterion for all separating sets

When every separating set — regular or irregular — must share one account, the condition tightens and reaches beyond the queried paths.

Theorem 4 (single-OD classification, static form)**.** *Let $\mathcal{G}$ be a connected multigraph with one OD pair (s,t), $s \neq t$, let $w \in \mathbb{R}^m$, and let $D = \mathrm{diag}(w)$. The* response-independent OD cut consistency property

$$\exists\, \alpha \in \mathbb{R}: \qquad BDx = \alpha\, \mu \quad \text{for every } x \in \mathbb{R}^m \text{ with } Bx = \mu, \tag{8}$$

with the scale α independent of the response x, holds ***if and only if***

(i) *w is constant within every non-bridge atom; and*
(ii) *all atoms on the s–t block path $\mathcal{P}(s,t)$ share one common value.*

In that case every conservative response induces the same boundary vector: $BDx = w^*\mu$ for every x with $Bx = \mu$, where $w^* := w|_{\bigcup \mathcal{P}(s,t)}$ is the common value of (ii) and $\alpha = w^*$ is the scale in (8).

Condition (i) concerns every atom, including those off the OD path. Their constants may differ; only the traversed atoms must share the scale w^*. Off-path atoms are constrained because a signed circulation can be added to any conservative response, and heterogeneous weights can turn it into node divergence. Connected regions change this conclusion, as Theorem 1 shows. The proof adds a virtual edge from t back to s (Appendix A.2): conservative OD responses then become circulations on the extended graph, so the boundary vector is response-independent exactly when diagonal weighting preserves the circulation space — equivalently, when every weighted cycle has zero node divergence — which forces equal weights within each cycle atom and one shared weight along the atoms the OD block path connects.

Theorem 5 (multi-OD classification, static form at a shared realized flow state)**.** *Let $\mathcal{G}$ be connected, let each OD pair satisfy $s_j \neq t_j$, and let $w \in \mathbb{R}^m$, $D = \mathrm{diag}(w)$. The response-independent OD cut consistency property (8) — for each queried j, some $\alpha_j \in \mathbb{R}$ with $BDx = \alpha_j \mu_j$ for every x with $Bx = \mu_j$ — holds simultaneously for the queried ODs* ***if and only if*** *w is constant on each OD-compatibility class $\bar{E}_k$.*

The classification concerns all signed responses compatible with vehicle conservation; an equilibrium model generally realizes only a subset of these responses (Appendix A.3). All queries use the same realized weight vector. Queries with $\Delta q_j = 0$ have zero attribution columns and need not enter the closure. Requiring all ordered pairs of distinct vertices collapses the classes to one per connected component (Corollary A.4).

4.4. Pathwise attribution and implications for learned assignment

Corollary 6 (path form: endpoint implication)**.** *Let $f \in C^1$ be conservative on an open set, $T \in C^1$, $\gamma(\lambda) = q^0 + \lambda\Delta q$ the straight-line path with $\Delta q \neq 0$, and $w(\lambda) = \nabla T(f(\gamma(\lambda)))$, $D_\lambda = \mathrm{diag}(w(\lambda))$, shared by all participating ODs. If the equivalent conditions of Theorems 4 and 5 hold at $w(\lambda)$ for every $\lambda \in [0,1]$, then, for every participating j simultaneously,*

$$B\widetilde{C}_{:,j} = \beta_j\, \mu_j$$

holds with

$$\beta_j = \Delta q_j \int_0^1 w^*_{k(j)}(\lambda)\, d\lambda,$$

where $w^*_{k(j)}$ is the common value on the compatibility class $k(j)$ containing the atoms of $\mathcal{P}_j$.

The specified edgewise outcome and the network representation are both part of the object. On two series links, an outcome whose marginal weights oscillate around unity can integrate to the same balanced column $(1,1)$ as the linear outcome on one demand segment, yet fail to balance on another (Example A.7); subdividing a link leaves the composite map unchanged but creates interior divergence and even halves the original cut value at the balanced split (Example A.8). Response independence concerns conservative responses on a fixed network; it does not make the account invariant to road subdivision.

Corollary 7 (conservative learned assignment)**.** *Let f^L be an exactly conservative learned assignment map on an open domain. Then its flow-valued integrated gradients inherit OD conservation, $BG^L_{:,j} = \mu_j \Delta q_j$. Write $w(\lambda) = \nabla T(f^L(\gamma(\lambda)))$ along the attribution path. For each λ, structural cut consistency of the learned explanations is equivalent to the weight conditions of Theorems 4 and 5 at $w(\lambda)$ for the unrestricted separating family, and to those of Theorem 1 for two-sided connected regions. When the unrestricted-family conditions hold at every λ, the integrated explanations are realized cut consistent, $B\widetilde{C}^L_{:,j} = \beta_j \mu_j$ (Corollary 6); when they fail at some λ, the node representation of Section 5 gives the exact spatial explanation range of the learned column over the unrestricted family and its exact maximum boundary disagreement from any reference assignment, with restricted families scanned through the same node sums.*

Proof. The conservation transfer is Eq. (4) applied to f^L. The pointwise equivalences are the two classification theorems applied to the fixed weight vector $w(\lambda)$; the integral claim is Corollary 6 with $D_\lambda = \mathrm{diag}(w(\lambda))$; the fallback is Corollary 9. □

Conservation of predicted flows is thus the first step of a chain whose later steps are topological; Section 7 evaluates the whole chain empirically.

5. Exact transferable summaries and certified boundary profiles

5.1. Obtaining attribution columns

For a separable outcome $T(F) = \sum_e g_e(F_e)$, a single changing OD gives

$$\widetilde{C}_{e,j} = g_e(F^1_e) - g_e(F^0_e). \tag{9}$$

Along this demand path, $dF_e/d\lambda = \Delta q_j \partial_j f_e$; the fundamental theorem of calculus applies link by link, including endogenous background rerouting. Two endpoint assignments therefore recover the attribution column without sensitivity integration. The boundary account is $H_U(F^1) - H_U(F^0)$. For jointly changing ODs, endpoint differences determine only the sum of attribution columns. Sensitivity integration in Eq. (2) separates the individual OD contributions. Both cases then use the same node representation and spatial aggregation below. After preprocessing, all accounts for a prescribed screenline family are obtained from one incidence multiplication and a linear scan; numerical comparisons with direct crossing sums agree to machine precision, and same-task timings against endpoint and vectorized alternatives are collected in Appendix B.6.

5.2. Node representation and exact account ranges

Once the classes are constructed, structural cut consistency over the unrestricted family of all separating regions is tested by

$$\eta_{\mathrm{all}}(\lambda) = \max_{k} \max_{e \in \bar{E}_k} |w_e(\lambda) - \bar{w}_k(\lambda)|, \qquad \bar{w}_k = |\bar{E}_k|^{-1} \sum_{e \in \bar{E}_k} w_e, \tag{10}$$

and the joint condition of Theorem 5 is exactly $\eta_{\mathrm{all}} = 0$. For two-sided connected regions, let $E_Q = \bigcup_j \bigcup_{A \in \mathcal{P}_j} A$ collect the edges of all queried block paths and define

$$\eta_{\mathrm{conn}}(\lambda) = \max_{k:\, \bar{E}_k \cap E_Q \neq \varnothing} \max_{e \in \bar{E}_k} |w_e(\lambda) - \bar{w}_k(\lambda)|, \tag{11}$$

which omits every class on no queried path. By Corollary 2, $\eta_{\mathrm{conn}} = 0$ is exactly the connected-region compatibility condition, and off-path atoms may remain internally heterogeneous, so $\eta_{\mathrm{conn}} = 0$ does not imply $\eta_{\mathrm{all}} = 0$. Class construction uses the block-cut tree and transitive merging; testing a weight vector then needs one edge scan. No fitted-model derivatives are required for a linear outcome.

For a realized integrated attribution, define

$$d_j = (I - P_{\mu_j}) B \widetilde{C}_{:,j}, \qquad P_{\mu_j} = \frac{\mu_j \mu_j^\top}{\|\mu_j\|_2^2}. \tag{12}$$

Endpoint cut consistency over all separating regions holds precisely when $d_j = 0$; only then does $\mu_j^\top B \widetilde{C}_{:,j} / \|\mu_j\|_2^2$ represent a common cut value, whatever mechanism produced the balance (Section 3.2). For a connected family or another prescribed finite family, $d_j = 0$ is sufficient but not necessary; realized consistency is then read from the account range over that family.

Without a connectivity restriction, each interior node can enter the separating set independently. Therefore Eq. (5) gives

$$S_j := \max_{U} \phi_j(U) - \min_{U} \phi_j(U) = \sum_{v \notin \{s_j, t_j\}} |(B \widetilde{C}_{:,j})_v|. \tag{13}$$

The extrema themselves have a closed form: writing $b_j = B \widetilde{C}_{:,j}$, $(x)_+ = \max(x, 0)$, and summing over interior nodes $v \notin \{s_j, t_j\}$,

$$\phi_j^{\min} = \phi_j(\{s_j\}) - \sum_v (b_{jv})_+, \qquad \phi_j^{\max} = \phi_j(\{s_j\}) + \sum_v (-b_{jv})_+, \tag{14}$$

since the minimizing set retains exactly the interior nodes with positive divergence. Subtracting gives, for every separating set U with $I_j = V \setminus \{s_j, t_j\}$,

$$\phi_j(U) - \phi_j^{\min} = \sum_{v \in I_j \setminus U} (b_{jv})_+ + \sum_{v \in I_j \cap U} (-b_{jv})_+, \tag{15}$$

so the gap to the unrestricted minimum is exactly the positive divergence the region omits plus the negative divergence it contains. When $\phi_j(\{s_j\}) > 0$, the unrestricted family therefore contains a negative account whenever the interior positive divergence $\sum_v (b_{jv})_+$ exceeds the origin-cut account: node imbalance, extremal location, and sign crossing are the same statement. $S_j = 0$ is equivalent to $d_j = 0$. The projector in Eq. (12) changes only the two endpoint entries. Cauchy–Schwarz on the $n - 2$ interior entries therefore gives

$$S_j = \sum_{v \notin \{s_j, t_j\}} |(d_j)_v| \leq \sqrt{n-2}\, \|d_j\|_2. \tag{16}$$

For network regions whose induced subgraphs and complements are both connected, define S_j^{conn} by restricting the same extrema to that family. Then $S_j^{\mathrm{conn}} \leq S_j$; a small restricted range alone does not establish the full structural criterion. We report both ranges as percentages of the absolute origin-cut value

Table 3
Structural tests and realized checks by admissible region family. E_Q is defined in Eq. (11); a small restricted range does not establish the unrestricted criterion.

Question	All separating regions	Connected / prescribed $\mathcal{F}$
Structural guarantee	$\eta_{\text{all}} = 0$	connected: $\eta_{\text{conn}} = 0$
Realized consistency	$d_j = 0$, equivalent to $S_j = 0$	account range over the family is zero
Maximum model difference	$\mathcal{K}_j(e_j)$, exact	upper bound; scan $\mathcal{F}$ for the exact value

$|\phi_j(\{s_j\})|$, which remains explicit because the ratio depends on that denominator (absolute ranges when it is zero).

Table 3 collects which test answers which question for each region family; the unrestricted column concerns all separating regions, and the second column concerns connected families or a prescribed finite family $\mathcal{F}$.

Corollary 8 (best common boundary value)**.** *For the unrestricted family, the unique best constant in maximum absolute account error is*

$$\beta_j^\star = \frac{\phi_j^{\max} + \phi_j^{\min}}{2} = \frac{(b_j)_{t_j} - (b_j)_{s_j}}{2}, \qquad \min_{\beta\in\mathbb{R}} \max_{U\in\mathcal{U}_j^{\text{all}}} |\phi_j(U) - \beta| = \frac{S_j}{2}.$$

Hence all separating accounts lie within a prescribed absolute tolerance ε of some common value if and only if $S_j \le 2\varepsilon$.

Proof. For any constant, its largest error at the two attained extrema is at least half their difference; their midpoint attains this bound for every account. Equation (14) and $\mathbf{1}^\top b_j = 0$ give the midpoint formula. □

The coefficient agrees with the endpoint projection in Eq. (12). It approximates the original accounts; it does not change their edge attribution or define a new system contribution.

For any nonempty prescribed finite family $\mathcal{F}$, the same midpoint argument gives

$$\beta_j^{\mathcal{F}} = \frac{\phi_{j,\max}^{\mathcal{F}} + \phi_{j,\min}^{\mathcal{F}}}{2}, \qquad \varepsilon_j^{\mathcal{F}} = \frac{\phi_{j,\max}^{\mathcal{F}} - \phi_{j,\min}^{\mathcal{F}}}{2}, \qquad |\phi_j(U) - \beta_j^{\mathcal{F}}| \le \varepsilon_j^{\mathcal{F}} \quad (U \in \mathcal{F}). \tag{17}$$

Every query therefore receives a usable output under a two-stage rule. With a specified relative tolerance τ, first compare $\varepsilon_j^{\mathcal{F}}$ with $\tau \max(|\phi_{j,\min}^{\mathcal{F}}|, |\phi_{j,\max}^{\mathcal{F}}|)$; if the family admits one common value within tolerance, report $\beta_j^{\mathcal{F}}$ with its exact error $\varepsilon_j^{\mathcal{F}}$; otherwise report the boundary response profile itself, or its minimal contiguous piecewise summary under the same tolerance (Proposition 10). The absolute pair remains available regardless of the normalization, and no query ends in a bare non-transferable verdict: the outcome is either a certified common value or a certified profile.

5.3. Exact scanning of predefined straight screenlines

Fix D directions before evaluating attribution. For each direction, sort nodes by their coordinate projection and group equal projection values; no screenline splits such a group. A candidate position is a gap between adjacent groups. Forward and reverse union–find sweeps record whether each prefix and suffix induces a connected subgraph. This geometry and topology preprocessing is reusable across queries, demand paths and models. For one attribution column, form $b_j = B\widetilde{C}_{:,j}$ and prefix sums $P_{j,k} = \sum_{i\le k}(b_j)_{\pi(i)}$. If s_j lies in the prefix and t_j outside, its account is $-P_{j,k}$; for the reverse orientation it is $P_{j,k}$ because $\mathbf{1}^\top b_j = 0$. Retain only endpoint-separating gaps with both sides connected. Scanning these values gives exact accounts and extrema over this prescribed finite screenline family. It does not optimize over every possible direction in the plane. Preprocessing takes $O(D(|V|\log|V| + |E|\alpha(|V|)))$; after preprocessing, a column takes $O(|E| + D|V|)$ using sparse incidence multiplication. Duplicate regions may be removed without changing extrema.

On Sioux Falls, 36 directions at five-degree intervals give 1642 admissible query–position accounts across six queries (duplicates across directions retained for all algorithms). On exactly these oriented regions, the prefix scan, a vectorized baseline that precomputes the signed crossing matrix $A_{\mathcal{F}}$ (one sparse row

Algorithm 1: Spatial accounts from an attribution column

Input: incidence B, query (s_j, t_j), edge attribution c_j, selected regions U, W, and optionally a comparison column c'_j.

0. For a single changing OD and separable outcome, obtain c_j from the two endpoint states using Eq. (9); for joint changes, input the separately integrated column.

1. Compute $b_j = Bc_j$; return its node distribution.

2. Return $\phi_j(U) = -\mathbf{1}_U^\top b_j$ and $\phi_j(U) - \phi_j(W) = -(\mathbf{1}_U - \mathbf{1}_W)^\top b_j$.

3. Scan interior nodes to return the extrema in Eq. (14) and the midpoint and half-range in Eq. (17), and their sets $\{s_j\} \cup \{v : b_{jv} > 0\}$ and $\{s_j\} \cup \{v : b_{jv} < 0\}$.

4. If c'_j is supplied, return $\mathcal{K}_j(c_j - c'_j)$ as the exact maximum absolute error over unrestricted separating regions.

The incidence product and scan cost $O(|E| + |V|)$ per column; selected-region sums cost $O(|V|)$. The same node representation serves all these tasks. The columns must already have been computed; their reference-attribution cost remains separate.

per oriented region, $+1$ leaving and -1 entering) and aggregates every account with one sparse product $A_{\mathcal{F}}C$, and direct crossing-edge summation all agree to 2.2×10^{-11} TSTT units once the reference columns are available. Appendix B.6 reports the same-task timings of the three aggregations and the amortization threshold of the precomputed matrix.

5.4. Exact boundary error and response supervision

For a query j and any edge vector h, define the seminorm

$$\mathcal{K}_j(h) = \frac{1}{2}\left(\left| (Bh)_{t_j} - (Bh)_{s_j} \right| + \sum_{v \notin \{s_j, t_j\}} |(Bh)_v| \right). \tag{18}$$

Corollary 9 (exact maximum boundary error)**.** *Let $e_j = \widetilde{C}^L_{:,j} - \widetilde{C}^S_{:,j}$ be the difference between learned and reference attribution columns. Then*

$$\max_{U : s_j \in U,\ t_j \notin U} |\phi^L_j(U) - \phi^S_j(U)| = \mathcal{K}_j(e_j). \tag{19}$$

At a common demand q, write $h_j(q) = D^L(q)J^L_j(q) - D^S(q)J^S_j(q)$. For any subfamily $\mathcal{U}_j$ of separating boundaries,

$$\epsilon^{\mathcal{U}}_j := \max_{U \in \mathcal{U}_j} |\phi^L_j(U) - \phi^S_j(U)| \le |\Delta q_j| \int_0^1 \mathcal{K}_j(h_j(\gamma(\lambda)))\, d\lambda. \tag{20}$$

The proof in Appendix A.5 applies Eq. (14) to the error column. Evaluation requires one incidence multiplication and node scan, without enumerating boundaries. Connected regions and straight screenlines inherit an upper bound; their restricted maxima need not equal $\mathcal{K}_j(e_j)$. The endpoint term must be retained: the two explanations may each be cut-consistent but have different common boundary values, which internal imbalance alone misses.

The seminorm evaluates spatial explanation errors directly. It also defines reference-matching losses, whose complete specifications are collected in Appendix B.8. Their comparison is an application of the spatial framework (Section 7).

Scope and approximate conservation. The classification fixes the road representation, outcome weights and admissible regions, and quantifies over all signed conservative responses. For a model with residual $r(q) = Bf(q) - Mq - b$ and constant b, the corresponding flow-attribution residual is

$$R_j := BG_{:,j} - \mu_j \Delta q_j = \Delta q_j \int_0^1 \partial_j r(\gamma(\lambda))\, d\lambda, \qquad \sum_j R_j = r(q^1) - r(q^0). \tag{21}$$

Small endpoint residuals control the sum, but individual columns can cancel. Approximate output conservation controls attribution residuals only when derivative regularity is also available; Appendix A.4 gives the $C^{1,1}$ conditions, a local bound, sharpness examples and an extension to weighted accounts. The traffic experiments below use exactly conservative models.

Table 4
Experiment map.

Ex.	Network and queries	Purpose	Region families
A	Sioux Falls, 8 largest ODs	background rerouting, spatial ranges	unrestricted, connected
B	Sioux Falls, 6 diverse ODs	signs, mechanisms, screenlines	connected, straight screenlines
C	Sioux Falls, 18 stratified ODs	common summaries, load levels, tolerance	fixed geometric family
D	Anaheim, 6 largest ODs	complete profiles, piecewise summaries	distance-progress family
E	learned models, 96 demand paths	explanation error, response supervision	geometric and full-boundary

6. Numerical evidence and transportation interpretation

Sioux Falls absolute accounts use benchmark flows in veh/h and time units of 0.01 h; multiply an account by 0.01 for the hourly vehicle-hour total. Anaheim uses vehicle-minutes per hour. All experiments use the stated fixed-path logit-SUE assignment models and prespecified demand scenarios; shares reported over boundary families are enumeration shares over those families. Table 4 maps the five experiments; the section answers three questions in order: when one boundary summary is adequate, what generates spatial variation, and what a complete spatial profile with compact summaries looks like.

6.1. When is one boundary summary adequate?

Sioux Falls has 24 nodes and 76 arcs in one cycle atom. Its structural weight condition is therefore the same for every OD; the experiments measure variation due to realized responses and heterogeneous marginal costs. The query-dependent classification is illustrated separately in Figure 1.

Six input-selected queries away from node 10 change jointly from 0.2 to 1.6 times their base values with all other demands fixed and routes responsive; they are the shortest and longest free-flow ODs in each demand tercile (Appendix B.5). Table 5 reports their ranges. The two-stage rule of Section 5 separates them sharply: the geometric accounts for $(7, 18)$ lie in $[642, 702]$, their midpoint 672 has maximum absolute error 30, or 4.3% of the largest account, and this family admits one common summary at a 5% tolerance; the $(1, 19)$ and $(7, 12)$ geometric ranges cross zero and require boundary-specific reporting at every tolerance considered below.

Table 5
System contributions and spatial ranges for six prespecified queries, setting C (TSTT units). Connected: all admissible connected regions; screenline: the geometric family constructed independently of attribution values. All C_j are positive. Origin-cut accounts, negative shares, and neighbourhood ranges are retained in Appendix C.5.

OD	C_j	Connected [min, max]	Screenline [min, max]
$(7, 18)$	672.9	[634, 710]	[642, 702]
$(1, 19)$	50,097.7	[−13,502, 24,260]	[−2,767, 22,124]
$(4, 5)$	2,826.0	[2,396, 3,452]	[2,652, 3,176]
$(1, 15)$	77,403.2	[−8,286, 40,540]	[511, 36,621]
$(6, 8)$	73,964.9	[35,307, 66,695]	[45,230, 65,286]
$(7, 12)$	117,482.3	[−17,616, 60,029]	[−2,967, 57,780]

Experiment C widens the view from six queries to an input-only selection of 18 of the 528 positive Sioux Falls OD demands, taking two lexicographically predetermined positions from each demand-by-free-flow-time tertile cell. Each OD changes alone over ±10% and ±20% intervals while other demands remain fixed with endogenous rerouting; the same 36-direction geometric family is used throughout, and Appendix C.9 reports every query and both widths. For each nonempty family, let $\beta = (\phi_{\max} + \phi_{\min})/2$ and $\varepsilon = (\phi_{\max} - \phi_{\min})/2$. This common summary minimizes the largest absolute discrepancy, so every reported boundary satisfies $|\phi(U) - \beta| \le \varepsilon$. The reported normalized error is

$$r_j^{\mathcal{F}} = 100\, \varepsilon_j^{\mathcal{F}} \Big/ \max\big(|\phi_{j,\min}^{\mathcal{F}}|, |\phi_{j,\max}^{\mathcal{F}}|\big), \tag{22}$$

Table 6
Common-summary examples at ±20%. The four rows show the two smallest and two largest normalized summary errors among the fixed 18 queries. Normalized error is Eq. (22); absolute values use the Sioux Falls cost units.

OD	System	Minimum	Maximum	β	ε	Norm. err. (%)
(12,3)	337.2	330.7	340.2	335.5	4.8	1.4
(4,3)	507.2	429.0	535.0	482.0	53.0	9.9
(18,14)	3422.3	150.8	1929.7	1040.2	889.5	46.1
(6,21)	4577.8	-3.4	2349.4	1173.0	1176.4	50.1

Table 7
Common-summary statistics across baseline load levels (18 queries, focal ±20%, same 36-direction family). Normalized error is Eq. (22); the ratio column is the median geometric-to-unrestricted spread ratio.

Level	Queries within 5%	Mixed signs	Median error (%)	Median ratio	Max. negative share
0.7	2	0	34.5	0.529	0.000
1.0	1	1	44.4	0.574	0.0067
1.3	0	3	45.2	0.563	0.0403

the absolute deviation of every account from the common value as a percentage of the family's largest account magnitude; it does not bound each account's own relative error. We apply a fixed descriptive tolerance $r_j^{\mathcal{F}} \leq 5\%$; this is a reporting choice rather than a transport welfare threshold. When all accounts in a family share one sign, $r_j^{\mathcal{F}} \leq 50\%$, so values near 50% indicate that the smallest magnitude is far below the largest, not that the accounts nearly agree. At ±10% and ±20%, respectively, 1 and 1 of 18 queries meet this tolerance, while 1 and 1 have accounts of both signs. The median geometric-to-unrestricted spread ratio is 0.574 at both widths.

The ±10% and ±20% summaries of the same 18 queries are close in normalized terms, so further subdividing the perturbation width adds little information. The congestion state of the baseline is the more informative axis. All 528 demands are therefore scaled jointly to 0.7, 1.0 and 1.3 times their published values; at each level the level equilibrium is solved once, the focal query alone moves by ±20%, other demands stay fixed with responsive routes, and the endpoint identity supplies the attribution column. The level-1.0 rows are the existing ±20% results; levels 0.7 and 1.3 add 74 equilibrium solves under the identical geometric family and masks. Table 7 shows a monotone pattern: under the light baseline, accounts are flatter across the family — the median normalized error falls to 34.5%, two queries satisfy the 5% tolerance and no family crosses zero — whereas the heavy baseline makes every query exceed the tolerance, three families mix signs and the negative share of $(6, 21)$ reaches 4%. Per-query normalized errors rise from the light to the heavy level for 17 of 18 queries (e.g. $(10, 12)$: 18.6 → 35.2 → 40.3; $(1, 10)$: 31.2 → 44.8 → 48.0). Across the load levels, light-load conditions admit transferable single summaries more often, and heavy-load conditions require boundary-specific reporting.

The dependence on the reporting tolerance itself is summarized by the frontier of Table 8. Relaxing the tolerance from 2.5% to 10% keeps the number of Sioux Falls queries admitting a single common value between zero and two at every load level (rising by at most one per level), while the median number of contiguous segments per query falls from 42–53 to 18–22 of a median 86.5 family regions; on Anaheim's ordered families the same relaxation raises the compression of the 83–211 accounts per query from 91.4% (6–21 segments) to 96.4% (3–12 segments). With the same segment count, equal-count partitions have median worst-segment error 2.7–10.4 times the tolerance on Sioux Falls and 4.7–18.8 times on Anaheim, so position-aware segmentation is what buys the error control, not the segment count alone.

The load-level trend has a theory-side counterpart in the node representation. For each of the 18 columns and three levels, the imbalance vector d_j of Eq. (12) has norm $\|d_j\|_2$ rising from a median of 1196.4 at level 0.7 to 4803.2 at 1.0 and 15763.3 at 1.3 — a 13.2-fold increase from the light to the heavy baseline — while the unrestricted range S_j rises from 3102.2 to 52766.2. Across all 54 query–level pairs, the rank correlation between $\|d_j\|_2$ and the geometric-family spread is 0.99 (Spearman; 0.99 for S_j), and

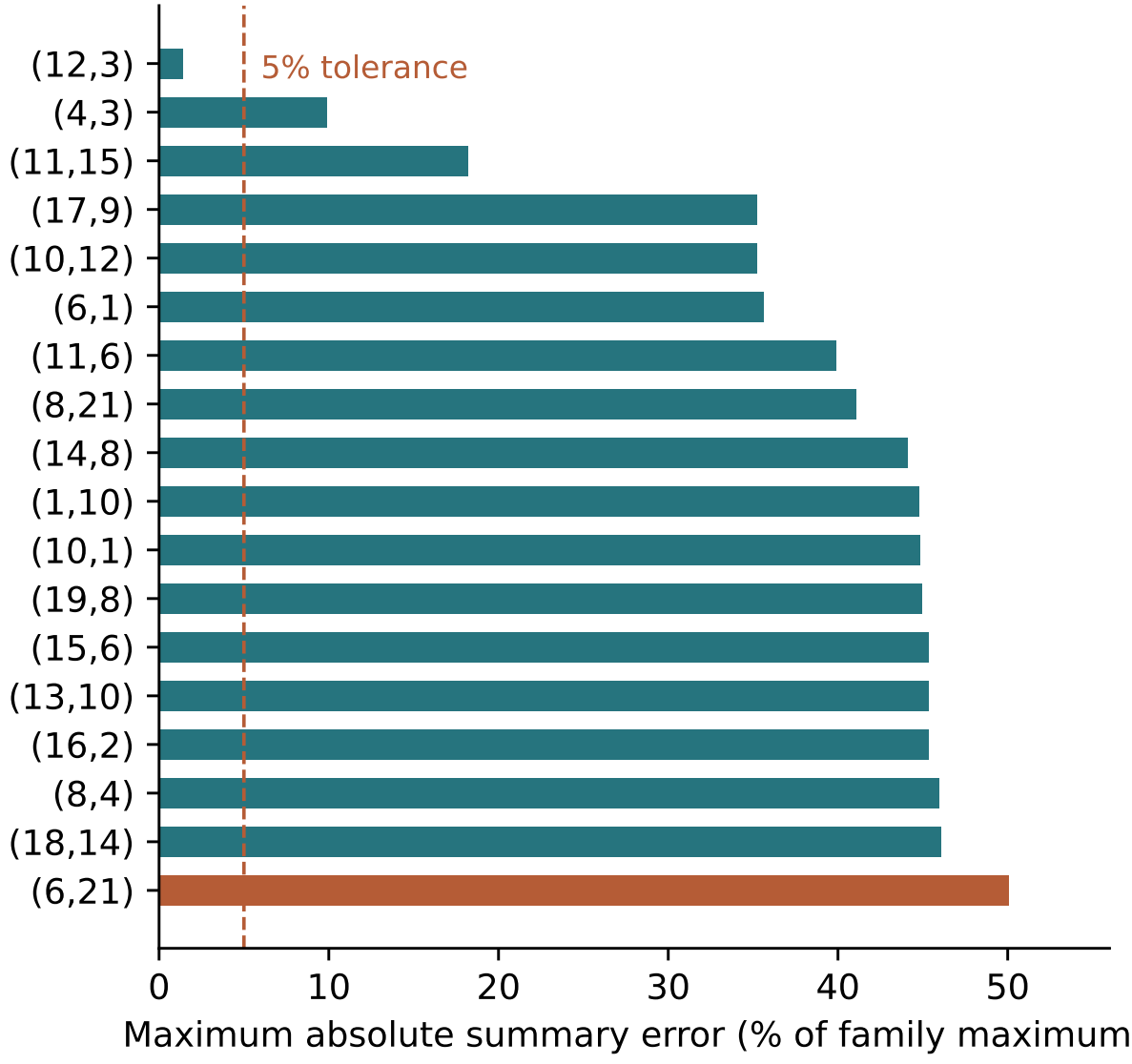


Figure 2: Maximum absolute summary error, as a percentage of each family's maximum account magnitude (Eq. (22)), over all 18 input-stratified queries at $\pm 20\%$, with the queries reordered by error. The tolerance is fixed at 5%; the complete account intervals appear in Figure C.3 in Appendix C.9.

the realized ratio $S_j/\|d_j\|_2$ has median 2.53 against the Cauchy–Schwarz bound $\sqrt{n-2} = 4.69$ of Eq. (16). The normalized summary error, whose denominator varies by query, correlates more moderately (0.53). The theory thus not only classifies when spatial ranges vanish; the load-level growth of the ranges is consistent with heavier congestion weighting amplifying the node imbalance that Eq. (13) sums into the spatial range — a mechanistic reading, since the load levels change the realized weights, responses and demand increments together rather than the weights alone.

6.2. What generates spatial variation?

Experiment A varies the background treatment. Eight largest published OD demands, all incident to node 10, change jointly from 0.2 to 1.6 times their base values; every study OD has ten fixed paths. Settings A, B and C use zero background, frozen background link flows and fixed background demands with endogenous rerouting, respectively. Setting C assigns all 528 positive demands (full protocols in Appendix B). Uniform

Table 8
Tolerance sensitivity of common summaries (experiment C for Sioux Falls, 18 queries per level; experiment D for Anaheim, six queries). Single: number of queries whose family admits one common value at the stated tolerance; segments: median (Sioux Falls, of a median 86.5 regions) or per-query range (Anaheim) of minimal contiguous segments; compression: 1−segments/regions.

Tolerance τ	Sioux single (0.7/1.0/1.3)	Sioux med. segments	Anaheim segments	Anaheim compression
2.5%	2/1/0	42.0/50.5/52.5	6–21	91.4%
5%	2/1/0	29.5/38.0/38.5	4–14	93.8%
7.5%	2/1/1	22.5/26.5/29.0	3–12	95.5%
10%	2/2/1	18.0/21.0/22.0	3–12	96.4%

weights provide a compatible control. The active-link outcome $T(F) = F_{10\to16}$ preserves the $(10, 16)$ direction but permits a response-dependent scale; TSTT produces internal imbalance. Figure C.1 illustrates these patterns. For its seed-zero TSTT response, expanding $\{10\}$ to $\{10, 17\}$ lowers the account from 5.77×10^4 to 4.34×10^4, approximately 92% of the unrestricted spread. The direct-SUE decrease is 24.80% of its own origin-cut account (Appendix C.1). Both sides of these regions are connected. Free-flow weights explain 84.7% of the node-17 imbalance: all incident links have free-flow time 2 except $10 \leftrightarrow 17$ with time 8, giving $[B\,\mathrm{diag}(t_0)G_j]_{17} = 6(G_{10\to17,j} - G_{17\to10,j})$.

Table 9 shows three representative queries of the eight under the three background settings (the complete table is in Appendix C.2); rerouting produces appreciable spatial variation for every query, and connected regions retain 60.41–76.85% of the unrestricted range across the full set. Normalizations use each setting's own origin account: for $(17, 10)$, the percentage falls from 59.95% in A to 54.77% in C while the absolute spread rises from 17,607.8 to 123,202.1, approximately sevenfold.

Table 9
TSTT cut ranges as percentages of each account's absolute origin-cut value; three representative queries shown, complete table in Appendix C.2. A: zero background; B: frozen background flows; C: background demands fixed, all traffic re-equilibrated. Learned values are five-seed medians (ranges); all SUE columns are direct (41 quadrature points, implicit sensitivities).

	Unrestricted range (%)				Connected (%)
OD	Learned A	SUE A	SUE B	SUE C	SUE C
$(17, 10)$	59.71 (58.92–61.23)	59.95	56.38	54.77	35.27
$(16, 10)$	36.80 (36.41–38.40)	36.86	39.99	166.26	127.72
$(11, 10)$	0.18 (0.03–0.27)	0.0073	0.79	81.09	48.98

Same-state response decomposition. Comparing A, B and C changes both costs and sensitivities. At each common setting-C state, split responses into study and background parts, with $BJ_j^{\mathrm{study}} = \mu_j$ and $BJ_j^{\mathrm{bg}} = 0$. Weight both by the same total-flow TSTT gradient, yielding

$$\widetilde{C}_j^{\mathrm{tot}} = \widetilde{C}_j^{\mathrm{study}} + \widetilde{C}_j^{\mathrm{bg}}. \tag{23}$$

At node 15 for $(10, 16)$, study and background divergence contributions are −973.1 and 49,103.4. Yet on the specified expansion $\{10\} \to \{10, 17\}$, the account changes only −3361.1 (−1305.0 study, −2056.1 background), or 1.23% of the initial account. Large extremal variation can coexist with similar selected accounts. Component differences add at a fixed region pair; their separate ranges do not add in general.

Free-flow and congestion mechanisms. For BPR costs, $g'_e = t_{0,e} + 0.75t_{0,e}(F_e/c_e)^4$. Crossing these two weight terms with the two response terms gives four additive contributions, evaluated at identical states and

the same reported boundary:

$$\phi_j(U_-) = \sum_{a\in\{\mathrm{ff},\mathrm{cong}\}} \sum_{b\in\{\mathrm{study},\mathrm{bg}\}} \phi_j^{a,b}, \qquad \phi_j^{a,b} = -\Delta q_j \int_0^1 \mathbf{1}_{U_-}^\top B D_\lambda^a J_j^b(\lambda)\, d\lambda.$$

Table 10 distinguishes the mechanisms behind the sign-crossing queries of Table 5: study congestion dominates $(1, 19)$, both congestion terms are negative for $(1, 15)$, and background congestion dominates $(7, 12)$. The combined free-flow contribution stays positive. The decomposition is additive at a common equilibrium state and identifies the contribution of each response–weight component. The sign pattern

Table 10
Four parts of the same boundary account at the reported U_- (setting C, six-query protocol; TSTT units; identical 41-point states, responses, and boundaries as Table 5). Study includes all six query ODs; background includes the other 522. Columns cross free-flow/congestion weights with study/background responses. Entries are rounded independently.

OD	Free-flow study	Free-flow background	Congestion study	Congestion background	Total
$(1, 19)$	−361.3	+759.0	−17,357.9	+3,458.6	−13,501.6
$(1, 15)$	+1,541.8	+333.8	−6,537.4	−3,624.2	−8,286.0
$(7, 12)$	+3,897.0	−1,311.1	+125.5	−20,327.6	−17,616.2

itself is a spatial property, not a magnitude artefact. Negative accounts occupy 17.8%, 2.5% and 7.3% of the 14,477, 15,191 and 16,943 connected regions of the three sign-crossing queries. For $(1, 19)$, the connected minimum −13,501.6 is attained on a 13-node region, whereas a 13-node free-flow neighbourhood has account +10,756.0: region size alone does not determine the account. At the minimizing boundary, outgoing attributions sum to 7,867.3 and incoming attributions to 21,368.8, so the signed crossing-link account is negative while $C_j = 50{,}097.7 > 0$; Figure C.2 maps all 26 crossing links. Figure 3 shows the same contrast on two straight screenlines of the prespecified geometric family: the minimum and maximum screenline accounts of $(1, 19)$ have opposite signs under one positive network-wide contribution. The straight-screenline family uses 36 directions at 5° intervals, gaps between node projections, two-sided connectivity and duplicate removal; construction is independent of attribution. It contains 23–152 distinct regions per query. The $(1, 19)$ and $(7, 12)$ negative minima occur on 2/152 and 10/148 screenlines. Origin-centred hop and free-flow neighbourhoods remain positive but can differ in magnitude by factors 8.9, 9.0 and 3.4 for the three sign-crossing queries (Appendix C.5). Sign stability and common-value accuracy are distinct properties.

6.3. Complete spatial profiles and compact summaries

The public Anaheim network has 416 nodes, 914 directed links and 1,406 positive OD demands (Transportation Networks for Research Core Team, 2024). Six largest demands change jointly from 0.8 to 1.2 times their published values; other demands remain fixed with endogenous route choices. Ten fixed free-flow shortest paths per OD exclude other zone centroids as intermediate nodes (first through node 39). This calculation uses the existing 41-state attribution columns. To obtain interpretable region locations without requiring coordinates, define free-flow progress $p(v) = [d_s(v) - d_t(v) + d_s(t)]/[2d_s(t)]$, using shortest distances in the underlying undirected network with the smaller free-flow time on parallel roads. The origin has progress zero and destination progress one. Group tied levels, retain prefixes with two connected induced sides that separate the OD, and orient every region toward the origin. This distance-based family depends on the network and query, before inspecting any attribution. It is a prescribed family, not all connected regions.

The case asks one transport question: how does the signed crossing-cost response of the demand change evolve along the origin–destination direction, and does the variation come mainly from the study traffic's congestion adjustment or from background rerouting? Figure 4 gives the answer for the query with the largest early–late difference: the account as a function of progress over the complete family, with the three prespecified positions marked, and the four-part same-state decomposition along the same axis. The profiles of the remaining five queries, which stay positive with family ranges spanning roughly one to one and a half orders of magnitude (largest-to-smallest account ratios 12.5–36.9), appear in Figure B.1 in Appendix B.7;

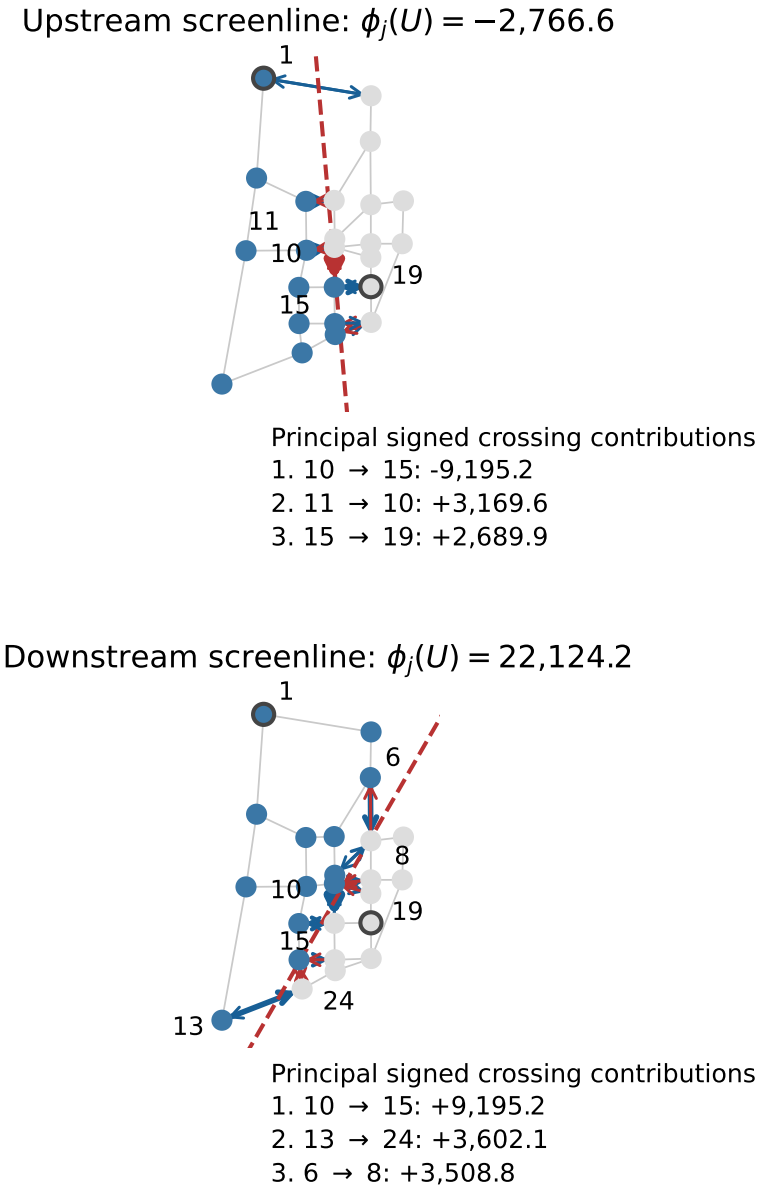


Figure 3: Contrasting screenline accounts for the same positive OD contribution. Two straight screenlines for OD $(1, 19)$, setting C. Blue nodes belong to U; grey nodes are outside; dashed lines separate the two sides. Node 1 is inside and node 19 outside in both panels. Upstream/downstream names the position of the separating line along the corridor: its crossing progress with the straight origin–destination segment is 0.61 (upper) versus 0.78 (lower). The two accounts have opposite signs while the network-wide contribution is positive. The lines are the minimum and maximum of the prespecified geometric family; the complete irregular connected minimum and its crossing-link decomposition appear in Figure C.2.

Table 11 reports the ranges and the exact progress levels and accounts of the three prespecified positions for all six queries. Query $(25, 2)$ falls from 801.5 at progress 0.250 to 412.0 at 0.500 and then rises to the family maximum 5751.2, which holds on the plateau from progress 0.724 to 0.818 before falling back to 2673.1 at 0.911. The decomposition panel locates the mechanism. At the early position the account is dominated by the free-flow study response (669.3 of the total 801.5, congestion study 111.5, background below 11); at the middle position the two study terms are comparable (211.5 free-flow against 212.6 congestion of the total 412.0, background below 11); at the late position the congestion study term reaches 5213.7 while both

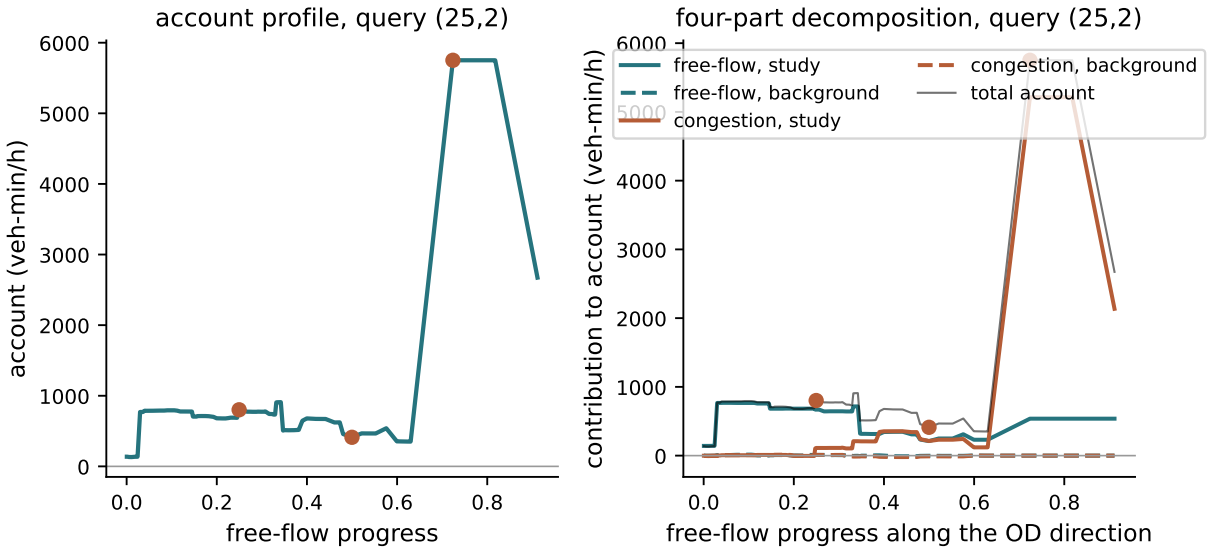


Figure 4: Anaheim spatial response profile for query $(25, 2)$ over the distance-based family. Left: the account versus free-flow progress along its own OD direction; ochre dots mark the three prespecified positions (nearest admissible gaps to progress 0.25, 0.50, 0.75). Right: the four-part same-state decomposition, crossing free-flow and congestion weights with study and background responses at identical states (teal: free-flow weights, ochre: congestion weights; solid: study response, dashed: background); the grey curve is the total account. Components close to 2×10^{-12} vehicle-minutes per hour.

Table 11
Anaheim accounts over the complete distance-based family, with the three prespecified positions (vehicle-minutes per hour). Column heads are the prespecified targets; the exact selected levels are (row order) $0.250/0.501/0.752$, $0.252/0.500/0.740$, $0.248/0.499/0.750$, $0.251/0.498/0.751$, $0.251/0.504/0.768$ and $0.250/0.500/0.724$.

OD	System	Unrestricted range	Family range	Regions	$\phi(0.25)$	$\phi(0.50)$	$\phi(0.75)$
(4,2)	34664.1	[-12712.6,20514.0]	[307.5,9832.1]	211	2323.5	1441.8	692.3
(1,2)	16773.6	[-6032.2,9762.0]	[172.7,6374.8]	83	259.0	818.0	360.5
(2,4)	12143.7	[-4374.6,6733.0]	[98.7,1905.9]	211	257.0	714.6	1061.8
(3,2)	14706.7	[-5613.8,9052.9]	[202.8,5777.4]	157	236.2	1087.3	434.6
(25,4)	6919.3	[-1805.3,3091.8]	[96.8,1212.2]	164	545.6	540.7	1212.2
(25,2)	11445.9	[-4157.3,6904.5]	[131.8,5751.2]	84	801.5	412.0	5751.2

background terms are below 0.1 in absolute value: the rise is congestion adjustment of the study traffic itself on destination-side links, not background rerouting. The decomposition is additive at a common equilibrium state and identifies the contribution of each response–weight component. Table 12 retains the early–late differences under the same decomposition at common-state, total-flow weights.

Because this family is totally ordered by progress, it also supports piecewise common summaries. The construction is stated for any ordered family.

Proposition 10 (minimal contiguous boundary summary)**.** *Fix an ordered family of boundary accounts $a_1, \ldots, a_N$ and a tolerance $\epsilon > 0$. Among partitions into contiguous intervals whose every interval has range at most 2ϵ and does not strictly mix signs, the longest-feasible-prefix procedure returns one with the minimum number of intervals.*

Proof. No feasible first interval can extend past the longest feasible prefix, so every optimal partition starts with a prefix no longer than the greedy choice; replacing the first interval of an optimal partition by the full greedy prefix cannot make the remainder infeasible, and the same argument applies inductively to the remainder. □

Table 12
Anaheim differences between preselected early and late regions, split by free-flow and congestion weights and study and background responses (vehicle-minutes per hour).

OD	Free-flow study	Free-flow background	Congestion study	Congestion background	Total
(4,2)	-416.91	-8.53	-1499.94	294.16	-1631.21
(1,2)	9.85	6.96	83.01	1.71	101.54
(2,4)	312.07	-65.41	667.06	-108.90	804.83
(3,2)	50.04	11.80	136.55	0.04	198.43
(25,4)	47.28	-40.11	666.28	-6.89	666.56
(25,2)	-131.77	-10.52	5102.23	-10.15	4949.79

Its value here is the coupling to the boundary scan, the node-difference interpretation, and a stated tolerance: a single interval reproduces the common summary of Eq. (17). On Anaheim, ordering by native progress, the 83–211 accounts per query reduce to 4–14 segments at the 5% tolerance (compression 90.4–95.2%), each within the stated tolerance by construction; the tolerance frontier is reported in Table 8. On Sioux Falls the mixed-direction family ordered by free-flow progress compresses less — the median query needs 38 segments for 86.5 regions — because distinct directions at nearby positions carry genuinely different accounts. Neither construction changes any original account; both summarize the same computed accounts with a stated worst error. The four components close within 2.0e-12 vehicle-minutes per hour; aggregation timings and the original construction costs appear in Appendix B.6.

7. Theory-guided evaluation and supervision of learned assignment

The preceding results identify weighted demand-response error as the quantity that governs boundary-explanation error. This section tests that implication on conservative learned assignment models. We compare models with similar flow accuracy but different response supervision and evaluate their explanations on the same fixed demand paths and region families. The same node representation compares learned explanations with reference equilibrium accounts. We evaluate flow prediction and attribution on the same 96 test demand paths, with twenty-four paths each containing one, two, three and six active queries. Path-level metrics average active queries and seeds before summarizing the paths; seed variability is reported separately. The evaluation asks which boundaries retain their account signs and magnitudes and whether discrepancies arise primarily from outcome weights or demand responses. Complete training objectives, validation selection and costs appear in Appendix B.9. Table 13 uses the common validation protocol, including update zero and every 100th continuation update. Each selected model satisfies the flow threshold; Appendix B.9 gives full objectives, denominators, the principal four-method comparison, and the earlier pointwise explorations. The test set was expanded from 24 to 96 demand paths for this evaluation — twenty-four paths per active-query group, generated by the same rule under a fresh seed — with the trained models, training data and validation selection frozen; paths are the independent unit, and active queries and seeds are averaged within a path before any summary. On the 96 test demand paths, response supervision lowers the median attribution-column error from 21.18% to 14.19% (a relative reduction of one third) and the geometric-boundary error from 8.74% to 5.75%, while both models retain flow-prediction errors near 0.03% (medians 0.0322% and 0.0292%). The improvement holds in every active-query group; between group medians, the six-query group has the largest absolute drop (8.24 percentage points) and the single-query group the largest relative reduction (39.5%). Path-level paired differences quantify the uncertainty: the median paired column-error difference is -5.03 percentage points with a path-level bootstrap 95% interval of $[-6.01, -4.48]$ (10,000 resamples), improving 92 of 96 paths; the paired geometric difference has median -2.06 with interval $[-2.76, -1.65]$, improving 90 of 96. Every group's bootstrap interval excludes zero, from $[-4.90, -2.95]$ for single-query paths to $[-8.59, -5.22]$ for six-query paths. Leave-one-path-out medians vary only within $[-5.11, -4.94]$ (column) and $[-2.11, -2.00]$ (geometric), so no single path carries the result. The system-contribution difference is secondary and opposite in sign: median $+0.45$ with interval $[0.01, 0.74]$, improving 39 of 96 paths. Response supervision improves the spatial metrics while both methods retain

Table 13
Prediction and spatial explanation on the same 96 test demand paths (24 per group). Entries are group medians of path-level values; flow errors are additionally averaged over the 41 states within each path. Improved: paths on which response supervision lowers the error.

Active	Column (%)		Geometric (%)		Improved	
queries	flow	resp.	flow	resp.	column	geometric
1	8.13	4.92	2.95	1.78	24/24	23/24
2	21.09	15.67	8.62	6.19	22/24	21/24
3	21.53	15.46	9.07	6.01	22/24	22/24
6	22.45	14.22	9.27	5.77	24/24	24/24
All 96	21.18	14.19	8.74	5.75	92/96	90/96

system-contribution errors below 3.5% (medians 2.52% and 3.13%); this separation confirms that scalar completeness and spatial explanation accuracy measure different properties.

On the original 24-path subset, the same-path decomposition in Eq. (C.1) identifies demand-response error, rather than marginal-cost weight error, as the larger component of the attribution discrepancy: for both models the response component has the larger norm on all 24 paths after averaging active queries and seeds, with path-median response and weight-component norms 256.1 and 4.1 under flow supervision and 158.1 and 3.8 under response supervision. Component norms need not add because signed components may cancel. Among the evaluated supervision objectives, response supervision produced the most consistent gains across active-query groups; the endpoint-edge objectives give no overall improvement and select the common initial model for three of five seeds (Appendix B.9).

Sign reliability is boundary specific. On the separate frozen six-query baseline segment of Appendix C.7, all five models retain negative accounts at the reference minimizing screenlines for $(1, 19)$ and $(7, 12)$. Their learned accounts lie in $[-3886.6, -3365.1]$ and $[-2810.4, -1916.6]$, respectively, against reference values -2766.6 and -2967.3. The largest errors over the prescribed screenline families, 1430.8 and 2499.2, are below the corresponding reference sign margins. These account-specific bounds establish the reported signs while retaining quantitative magnitude errors. This frozen-segment evidence is separate from the 96-path comparison above.

8. Conclusions

Spatial OD attribution admits an exact topological organization. For two-sided connected regions, a common boundary account is guaranteed precisely when marginal weights agree on the block atoms traversed by the queried OD paths; multiple queries merge these atoms transitively. When exact consistency is unavailable, node divergence yields the exact account range, the minimax common value and its error, and the maximum disagreement between learned and reference explanations. The Sioux Falls and Anaheim applications show how study traffic, background rerouting, free-flow costs and congestion jointly shape spatial response profiles, while compact summaries retain explicit accuracy guarantees. For conservative learned assignment, response supervision improves the spatial explanation metrics predicted by the theory. Together, these results turn spatial attribution from an ad hoc boundary aggregation into a transport-specific, mathematically controlled explanation framework.

Limitations. The guarantees concern a fixed road representation, outcome definition and admissible region family. Numerical conclusions concern the stated fixed-path assignment models and prespecified demand scenarios; subdividing roads can change crossing-link accounts, and broader demand domains require corresponding validation.

Code and data availability

Code, experiment configurations, numerical arrays, and scripts reproducing every table and figure form a versioned archive, with a README mapping each reported result to its generating script, fixed random

seeds, software versions, and checksums for the public TNTP inputs; the archive and the trained checkpoints are deposited for public release with the submission. The Sioux Falls and Anaheim networks and demands are available from the public Sioux Falls TNTP repository and Anaheim TNTP repository (Transportation Networks for Research Core Team, 2024); training targets are synthetic fixed-path logit-SUE assignments.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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A. Mathematical foundations and proofs

A.1. Conventions and conservation transfer

The incidence matrix $B \in \mathbb{R}^{n\times m}$ has entries -1 at an arc's origin and $+1$ at its destination; $\mu_j = e_{t_j} - e_{s_j}$ and $M = [\mu_1, \dots, \mu_K]$. Write $Z = \ker B$ and let $B^\dagger$ be the Moore–Penrose pseudoinverse. All norms below are Euclidean or their induced operator norms unless indicated otherwise. Graphs are loopless multigraphs: parallel and reciprocal arcs remain distinct edges, including in two-edge cycles. Cycle atoms and the block-cut tree are taken on the underlying undirected multigraph. Formally, $\iota(v)$ is the cut-vertex node when v is an articulation vertex and otherwise its unique block node; $\mathcal{P}(s,t)$ lists the atoms on the unique tree path from $\iota(s)$ to $\iota(t)$. A signed simple s–t path vector p satisfies $Bp = \mu$ and the full solution set is $p + Z$. Such a path uses only atoms of $\mathcal{P}(s,t)$ and meets every atom on it.

Signed responses. The circulation and response vectors are signed first-order sensitivities, not nonnegative commodity flows. At a strictly positive flow, sufficiently small signed circulation perturbations also preserve nonnegativity. A restricted response family is treated explicitly below.

Corollary A.1 (IG exchange)**.** *For an exactly conservative C^1 field on an open neighborhood of the attribution path, $BG_{:,j} = \mu_j \Delta q_j$.*

Proof. Differentiate $Bf(q) = Mq$ to obtain $BJ_f(q) = M$, then integrate the jth column along $\gamma(\lambda) = q^0 + \lambda\Delta q$.

A.2. Cycle structure and the classification proofs

Lemma A.2 (structure of circuit classes)**.** *The non-bridge atoms are the non-singleton circuit classes; each bridge is a singleton class. Moreover $Z = \bigoplus_A Z_A$, where $Z_A = \{z \in Z : \operatorname{supp} z \subseteq A\}$, and the summands have disjoint edge supports.*

Proof. No simple cycle crosses an articulation vertex into another block: it would revisit that vertex. Within a non-bridge block, any two edges lie on a common cycle. One way to see this is to subdivide the two edges; the resulting block is still biconnected, so its two new vertices lie on a common cycle, which descends to the required cycle. This is the usual circuit-class characterization of the graphic matroid (Oxley, 2011). Every signed circulation is a linear combination of signed simple-cycle vectors: subtract the fundamental cycles of a spanning forest to leave a circulation on a forest, necessarily zero by successive leaf elimination. Grouping these cycles by their blocks gives the direct sum, and distinct blocks have disjoint edge sets.

Lemma A.3 (circulation-preserving diagonal maps)**.** *For $D = \mathrm{diag}(w)$, $DZ \subseteq Z$ if and only if w is constant on every non-bridge atom; bridge weights are free.*

Proof. For any signed simple-cycle vector z_C, a circulation supported on that cycle has equal signed amplitudes on successive edges, by node balance. Thus $BDz_C = 0$ forces w constant on C, and Lemma A.2 propagates equality throughout each atom. Conversely, atomwise constancy makes D a scalar on every Z_A, so it preserves their direct sum. Bridges carry no circulation.

Proof (Theorem 4). Add a virtual reverse OD edge (t, s) and set

$$\bar{B} = [B, -\mu], \qquad \bar{D} = \mathrm{diag}(w, \alpha).$$

For a fixed α, property (8) is equivalent to $\bar{D} \ker \bar{B} \subseteq \ker \bar{B}$. Indeed, a kernel vector (x, a) obeys $Bx = a\mu$. If $a \neq 0$, apply (8) to x/a to obtain $BDx = a\alpha\mu$. If $a = 0$, subtract the identities for $p + x$ and p, with any fixed $Bp = \mu$, to obtain $BDx = 0$. These are exactly the two cases of $\bar{B}\bar{D}(x, a) = 0$. Conversely, applying kernel invariance to $(x, 1)$ gives (8) for every $Bx = \mu$, at that same α.

By Lemma A.3, this invariance holds precisely when the augmented weights are constant on its cycle atoms. The virtual edge and a simple s–t path form a cycle meeting every original atom on $\mathcal{P}(s, t)$, so these atoms merge with the virtual edge into one circuit class. No other atoms join it: every new simple cycle contains the virtual edge and a simple s–t path, which uses only those atoms. Other original atoms retain their classes. Thus the augmented criterion is exactly within-atom constancy everywhere and common value α on $\bigcup \mathcal{P}(s, t)$. This proves both directions and identifies $\alpha = w^*$ without changing the universal response family.

Proof (Theorem 5 and Corollary 6). Apply the single-OD theorem to each query at the same w. Necessity requires equality within every atom and among all atoms co-traversed by a queried block path. Their transitive closure is exactly the compatibility-class partition. Conversely, class constancy supplies both single-OD conditions for every query. Along the attribution path, exact conservation gives $BJ_f(\gamma(\lambda))\mathbf{e}_j = \mu_j$; the static theorem therefore yields

$$B\widetilde{C}_{:,j} = \Delta q_j \int_0^1 BD_\lambda J_f(\gamma(\lambda))\mathbf{e}_j \, d\lambda = \Delta q_j \left(\int_0^1 w^*_{k(j)}(\lambda) \, d\lambda \right) \mu_j.$$

Corollary A.4 (all-OD class collapse)**.** *Requiring the criterion at one common w for every ordered pair of distinct vertices makes all atoms in a connected component share one constant.*

Proof. For two atoms incident to the same cut vertex, choose endpoints on their opposite sides. Their OD block path contains both atoms, forcing their constants to agree. Connectivity of the block-cut tree propagates the equality throughout the component.

Remark A.1 (a common flow realization). The merging proof requires one common weight vector. Computing the queries at different realized flows does not in general supply this premise.

The calculation uses the minimum-norm solution of $Bx = \mu$ and checks $BDZ = 0$ on a basis of $\ker B$, which establishes response independence of the displayed boundary vector. The theorem supplies the uniform common-account guarantee in stages one and three.

Table A.1
Constructive query-dependent compatibility on Figure 1 of the main text. Ranges use all 512 separating sets per query and a conservative unit response; the weighted circulation residual is below 5×10^{-16} in all stages, so these ranges are independent of that response. The intentionally chosen linear weights illustrate compatibility; they are not the marginal costs of the original TSTT outcome.

Stage	$w_{A,b_1,B}$	$w_{b_2,C}$	w_D	Queried ODs	Account ranges
Initial query	1	2	3	(s,u)	$[1,1]$
Add query	1	2	3	$(s,u);(u,t)$	$[1,1];[1,2]$
Adjust merged class	1	1	3	$(s,u);(u,t)$	$[1,1];[1,1]$

A.3. Direction, integration, and network representation

At fixed weights $D = \operatorname{diag}(w)$, a realized model balances when $BDJ_j = \alpha_j \mu_j$ for its actual response. *Universal directional balance* requires $BDx \in \operatorname{span}\{\mu_j\}$ for every $Bx = \mu_j$, allowing its coefficient to depend on x; *structural cut consistency* additionally requires one common coefficient across all such responses, and is the property classified in the main text. For non-collinear queries with shared weights, universal directional balance already implies structural cut consistency because $BD \ker B \subseteq \bigcap_j \operatorname{span}\{\mu_j\} = \{0\}$; a query and its reverse alone do not suffice.

Proposition A.5 (universal directional criterion)**.** *Let $\hat{x}_j = B^\dagger \mu_j$, let N span Z with orthonormal columns, and let P_{μ_j} project orthogonally onto* $\operatorname{span}\{\mu_j\}$*. At fixed w, universal directional balance is equivalent to*

$$(I - P_{\mu_j})BD\hat{x}_j = 0, \qquad (I - P_{\mu_j})BDN = 0.$$

The scale may depend on the response. Requiring one common boundary value additionally requires $BDN = 0$. If directional balance holds along the whole path for a conservative model, its integrated column balances.

Proof. Every response is $\hat{x}_j + z$, $z \in Z$. Apply $I - P_{\mu_j}$ to $BD(\hat{x}_j + z)$ and vary z. The scale is $\mu_j^\top BD(\hat{x}_j + z)/\|\mu_j\|^2$ and is independent of z precisely when $BDZ = 0$. Integrating the realized directional identity gives the endpoint conclusion. Equivalently, invariance on all incidence fibers, $Bx = By \Rightarrow BDx = BDy$, is exactly $DZ \subseteq Z$, since $x - y$ ranges over Z.

Example A.6 (strict separation of the criteria)**.** *On the diamond $(s,a),(a,t),(s,b),(b,t)$ with weights $(1,1,1,2)$, the upper-path response $(1,1,0,0)$ has divergence μ after weighting, but the lower-path response has weighted divergence $(-1,0,-1,2)$ in node order (s,a,b,t), outside* $\operatorname{span}\{\mu\}$*. On the triangle $(s,t),(t,u),(u,s)$ with weights $(1,0,0)$, every response $x = (1+c,c,c)$ satisfies $BDx = (1+c)\mu$; direction balances, but the common boundary value fails because the circulation $(1,1,1)$ maps to $\mu \neq 0$.*

If directional balance holds for all queries at common w, differences give $BDZ \subseteq \bigcap_j \operatorname{span}\{\mu_j\}$. For non-collinear queries this intersection is $\{0\}$, so directional balance implies the common-value property. This uses the full response space Z.

Restricted families. For $S \subseteq Z$ with basis N_S, test responses in $x_j^{\mathrm{ref}} + S$, where $Bx_j^{\mathrm{ref}} = \mu_j$. The same proof gives the direction tests with $\hat{x}_j, N$ replaced by x_j^{ref}, N_S; common boundary value replaces the second test by $BDN_S = 0$. Using $\hat{x}_j$ requires $\hat{x}_j - x_j^{\mathrm{ref}} \in S$. For a fixed-path model $f = \Delta h$, $A_{\mathrm{OD}} h(q) = q$ and $B\Delta = MA_{\mathrm{OD}}$. Choose a path vector a_j with $A_{\mathrm{OD}} a_j = \mathbf{e}_j$ and set $x_j^{\mathrm{ref}} = \Delta a_j$. Differentiation yields

$$J_f \mathbf{e}_j - x_j^{\mathrm{ref}} = \Delta(\partial_j h - a_j) \in \Delta \ker A_{\mathrm{OD}} =: T_{\mathrm{swap}}.$$

This space includes OD path exchanges of every OD: background ODs may reroute even when their demands remain fixed.

Example A.7 (endpoint cancellation)**.** *For two series bridges, $f(q) = (q,q)$ and $T(F) = F_1 + F_2 + [\sin(2\pi F_1) - \sin(2\pi F_2)]/(2\pi)$ give*

$$\widetilde{C}_1 - \widetilde{C}_2 = \frac{\sin(2\pi q^1) - \sin(2\pi q^0)}{\pi}.$$

For $q^0 = 0$, the nonzero balanced endpoints in $[0,1]$ are $q^1 = 1/2, 1$, although the two marginal weights differ almost everywhere. At $q^1 = 0.3$ balance fails. Endpoint cancellation therefore does not establish the pointwise classification conditions.

Example A.8 (network subdivision)**.** *A single edge with $f(q) = q$, $T(F) = \tau F$, $\tau > 0$, has cut account $\tau\Delta q$. Subdivide it into $(s,v),(v,t)$ and set $f'(q) = (q,q)$, $T'(F_1,F_2) = \tau[\alpha F_1 + (1-\alpha)F_2]$, $0 < \alpha < 1$. The composite demand-to-outcome map is unchanged, but $\widetilde{C}' = \tau\Delta q(\alpha, 1-\alpha)$ and $(B\widetilde{C}')_v = \tau(2\alpha-1)\Delta q$. The two cuts account $\tau\alpha\Delta q$ and $\tau(1-\alpha)\Delta q$. They agree only for $\alpha = 1/2$, when each is half the undivided account; their sum remains the scalar contribution $\tau\Delta q$. The invariance studied here thus concerns conservative responses on a fixed network with a fixed edgewise outcome definition, rather than invariance under subdivision.*

A.4. Approximate conservation and attribution balance

The residual identity (21) shows that output residuals alone do not control attribution columns. For unit $v \in \operatorname{im} B$ and $r_\varepsilon(q) = \varepsilon v \tanh((q_1 - q_2)/\varepsilon)$, the field $f_\varepsilon(q) = B^\dagger(Mq + r_\varepsilon(q))$ has residual exactly r_ε, globally bounded by ε and zero on $q_1 = q_2$; yet on $q(\lambda) = \lambda(\delta,\delta)$ its partial derivatives are $v, -v$, so $R_1 = \delta v$ and $R_2 = -\delta v$, and its residual Jacobian Lipschitz constant grows as ε^{-1}. Added to an exactly conservative field with minimum flow $\varrho > 0$ on a compact positive-demand box, it preserves strictly positive flows when $\varepsilon\|B^\dagger\|_2 < \varrho$. The traffic experiments use exactly conservative models; these bounds delimit scope rather than enter the numerics.

Theorem A.9 (local neighborhood bound)**.** *Let $r \in C^{1,1}(\mathcal{N};\mathbb{R}^n)$ on an open neighborhood of the attribution path, with $\operatorname{Lip}(J_r) \le H$. Suppose, for some $\rho > 0$ and every j, λ, the coordinate segment $\gamma(\lambda) + [-\rho,\rho]\mathbf{e}_j$ lies in $\mathcal{N}$ and its two endpoint residual norms are at most ε. Then*

$$\|R_j\| \le |\Delta q_j|(\varepsilon/\rho + H\rho/2).$$

Proof. At $q = \gamma(\lambda)$, Taylor's formula gives $r(q \pm \rho\mathbf{e}_j) = r(q) \pm \rho\partial_j r(q) + \xi_\pm$ with $\|\xi_\pm\| \le H\rho^2/2$. Subtraction bounds $\|\partial_j r(q)\|$ by $\varepsilon/\rho + H\rho/2$; integration gives the claim.

The optimized square-root rate is sharp. Indeed $r_\varepsilon = \varepsilon v \sin((q_1 - q_2)/\sqrt{\varepsilon})$ has supremum norm ε and $\operatorname{Lip}(J_{r_\varepsilon}) \le 2$, while the same diagonal path gives $R_1 = \delta\sqrt{\varepsilon}v = -R_2$. Thus a uniform $o(\sqrt{\varepsilon})$ rate is impossible for bounded Jacobian Lipschitz constant.

Lemma A.10 (minimum-norm support)**.** *For a signed s–t path p, $\hat{x} = B^\dagger\mu = (I - P_Z)p$ is supported on $\bigcup \mathcal{P}(s,t)$.*

Proof. The minimum-norm point of $p + Z$ is its orthogonal projection onto $Z^\perp$. Lemma A.2 gives $P_Z = \sum_A P_{Z_A}$ with disjoint supports; both p and every $P_{Z_A}p$ vanish on off-path atoms.

Proposition A.11 (combined dispersion–residual bound)**.** *Let $f \in C^1$ near the attribution path, $r = Bf - Mq - b$ with the constant background balance b of Section 5.4 ($b = 0$ without background), and let $w(\lambda) = \nabla T(f(\gamma(\lambda)))$ be continuous. Let $\bar{w}(\lambda)$ be the compatibility-class mean vector and $\eta(\lambda) = \|w(\lambda) - \bar{w}(\lambda)\|_\infty$. Writing $J_j(\lambda) = J_f(\gamma(\lambda))\mathbf{e}_j$ and $d_j = (I - P_{\mu_j})B\widetilde{C}_{:,j}$, we have*

$$\|d_j\|_2 \le |\Delta q_j|\|B\|_2 \int_0^1 \left[\eta(\lambda)\|J_j(\lambda)\|_2 + \|\bar{w}(\lambda)\|_\infty\|B^\dagger\|_2\|\partial_j r(\gamma(\lambda))\|_2\right] d\lambda.$$

Proof. Differentiation gives $BJ_j = \mu_j + \partial_j r$ — b is constant and drops out of the derivative — so $J_j = \hat{x}_j + z_j + B^\dagger\partial_j r$, $z_j \in Z$. Let $\bar{D} = \operatorname{diag}(\bar{w})$. Class constancy gives $B\bar{D}z_j = 0$ by Lemma A.3. By Lemma A.10, $\hat{x}_j$ is supported on the OD block path, whose atoms lie in one compatibility class $k(j)$; hence $B\bar{D}\hat{x}_j = \bar{w}_{k(j)}\mu_j$. Therefore

$$(I - P_{\mu_j})BDJ_j = (I - P_{\mu_j})B\left[\operatorname{diag}(w - \bar{w})J_j + \bar{D}B^\dagger\partial_j r\right].$$

The projector is non-expansive, the two diagonal operator norms are η and $\|\bar{w}\|_\infty$, and integration gives the bound. For exact conservation only the dispersion term remains; for $\eta = 0$ only the derivative residual remains. Theorem A.9 can bound that derivative term under its local hypotheses.

A.5. Exact boundary error and its path bound

Proof. Set $b = Be_j$ and $I = V \setminus \{s_j, t_j\}$. Since $\mathbf{1}^\top b = 0$, Eq. (14) applied to e_j gives an interval of extrema with midpoint

$$-b_{s_j} - \frac{1}{2}\sum_{v\in I} b_v = \frac{1}{2}(b_{t_j} - b_{s_j})$$

and half-width $\frac{1}{2}\sum_{v\in I}|b_v|$. The larger absolute endpoint is the absolute midpoint plus this half-width, proving Eq. (19). This argument concerns the attained extrema; it does not require every value between them to be attained. The map $\mathcal{K}_j$ is a seminorm because it is a sum of absolute linear functionals. Moreover $e_j = \Delta q_j \int_0^1 h_j(\gamma(\lambda))\, d\lambda$. Its homogeneity and triangle inequality therefore yield

$$\epsilon_j^{\mathcal{U}} \le \mathcal{K}_j(e_j) \le |\Delta q_j| \int_0^1 \mathcal{K}_j(h_j(\gamma(\lambda)))\, d\lambda,$$

proving the path statement. Its kernel is $\ker B$: if the seminorm vanishes, all interior divergences vanish and $b_{s_j} = b_{t_j}$; the zero sum then makes both endpoints zero.

A.6. Connected-region classification

We retain the loopless multigraph conventions of Appendix A.1; reciprocal arcs are distinct undirected edges when defining atoms and connectivity.

Lemma A.12 (localization of a separating bond)**.** *If $G[U]$ and $G[V \setminus U]$ are connected, all edges of $\delta(U)$ lie in one cycle atom (or its singleton bridge). If $s \in U$ and $t \notin U$, that atom lies on $\mathcal{P}(s,t)$.*

Proof. For two distinct crossing edges, connect their endpoints by a simple path inside U and by another inside its complement. The two paths have disjoint vertex sets; together with the crossing edges they form a simple cycle, including the possible two-edge cycle. Hence all crossing edges share a circuit class. A singleton crossing edge is a bridge. A simple s–t path crosses this boundary, so it meets its atom; every atom met by such a path lies on $\mathcal{P}(s,t)$.

Lemma A.13 (connected prefixes in an atom)**.** *Let A be a non-bridge atom and let $a \neq b$ be two of its vertices. Its vertices admit an ordering $v_1 = a, \ldots, v_k = b$ such that every proper nonempty prefix and its suffix induce connected subgraphs. The assertion includes two-vertex parallel-edge atoms.*

Proof. For two vertices the assertion is immediate. Otherwise add an a–b edge temporarily if needed. The connected-prefix property is the classical st-numbering property (Even and Tarjan, 1976); we give a construction here. A biconnected graph with this edge admits an open-ear construction starting from a cycle containing that edge. For completeness, this construction follows directly from vertex connectivity. If vertices remain outside the current subgraph, any component of those vertices has at least two distinct attachment vertices in the current subgraph: a unique attachment would be an articulation of the original graph. A path through the component between two attachments supplies the next ear. Repeat until all vertices have entered. Edges left over are irrelevant to the ordering. The initial cycle exists because the added edge is not a bridge.

Order the other arc of the initial cycle from a to b. For each subsequent ear, let u precede v among its two attachment vertices; insert its internal vertices, in their order from u to v, immediately after u. Every new internal vertex has an earlier and a later neighbor. Existing vertices keep these neighbors. The first and last vertices remain a, b. Every vertex except a therefore has an earlier neighbor and every vertex except b a later neighbor. Repeated steps along these neighbors connect every prefix to a and every suffix to b. Removing the temporary edge does not affect this argument: it is never internal to a proper prefix or suffix.

Lemma A.14 (external components of a traversed atom)**.** *Let A be an atom of a connected loopless multigraph. Delete all edges of A while keeping its vertices. The remaining graph has exactly one connected component T_v for each $v \in V(A)$, containing v and no other vertex of A, and these components partition V. This holds with parallel edges and reciprocal arcs; for a bridge atom the components are the two singleton sides of the bridge.*

PROOF. No two distinct vertices $u, v \in V(A)$ are joined by a path that avoids the edges of A. Suppose such a path P existed. For a bridge atom $\{e\}$, the path P together with e would contain a simple cycle through the bridge e, but a bridge lies on no cycle. For a non-bridge atom (a block), pick any u–v path Q inside A; it uses only edges of A, possibly the single edge, or the two-edge cycle formed by parallel or reciprocal arcs. Walking along P from u until its first vertex w that lies on Q and returning to u along Q exhibits a simple cycle containing at least one edge of $Q \subseteq A$ and at least one edge of P outside A. Every simple cycle is contained in one block, so this cycle's block would contain edges of A and edges outside A, contradicting that a block's edge set is exactly the non-bridge atom A. Hence each $v \in V(A)$ reaches no other vertex of A without using A's edges, so the components $\{T_v\}_{v \in V(A)}$ are pairwise distinct; vertices outside A attach to exactly one of them, again because any path from such a vertex into A ends at a first vertex of A, and two different endpoints would contradict the claim just proved. The T_v therefore partition V.

PROOF (THEOREM 1). For sufficiency, Lemma A.12 places every crossing edge in one traversed atom. Its weight is α, so $-\mathbf{1}_U^\top BDx = -\alpha \mathbf{1}_U^\top Bx = -\alpha \mathbf{1}_U^\top (e_t - e_s) = \alpha$. No other edge enters this boundary sum.

For necessity, fix a traversed atom A, delete its edges, and form the components T_v of Lemma A.14. There are distinct vertices a, b with $s \in T_a$ and $t \in T_b$; these are the entry and exit of the OD block path in A. For a bridge atom, take the cut consisting of its s-side component. Since its unit conservative crossing response is fixed, the assumed account immediately gives that bridge's weight α.

For a non-bridge atom, choose the ordering of Lemma A.13. Set $U_i = \bigcup_{\ell \le i} T_{v_\ell}$ for $1 \le i < k$. The bags and the prefix/suffix connectivity show $U_i \in \mathcal{U}_{st}^{\mathrm{conn}}$. Its crossing edges are exactly those of the corresponding prefix in A. Now choose any signed local response y with $B_A y = e_b - e_a$, where B_A is the incidence matrix on A. Extend it by a signed unit path from s to a inside T_a and from b to t inside T_b, with zero on other edges. The result satisfies $Bx = \mu$. Thus the assumption gives

$$-\sum_{\ell \le i} (B_A D_A y)_{v_\ell} = \alpha \quad (1 \le i < k).$$

The first prefix gives $(B_A D_A y)_a = -\alpha$, differences of consecutive prefixes give zero at every interior vertex, and zero total incidence sum gives $(B_A D_A y)_b = \alpha$. Consequently $B_A D_A y = \alpha(e_b - e_a)$ for every local conservative y. The single-OD classification (Theorem 4), applied to the one-atom graph A, forces every edge in A to have weight α. This holds for every traversed atom. The sufficiency argument already shows that no off-path weight is constrained.

PROOF (COROLLARY 2). Apply Theorem 1 at the common w to each query. Each queried path requires one weight on all its edges; intersecting paths propagate equality transitively. Conversely these transitive equalities satisfy every single-query condition. Atoms not on any queried path enter none of these equalities.

A.7. Finite-change supervision and joint-path implication

PROOF (PROPOSITION B.1). Write $r \in \{L, S\}$ and $c^r(q) = g(f^r(q))$ componentwise. Along a single-coordinate segment $q(u) = q^- + u(q^+ - q^-)$, the chain rule gives

$$\frac{d}{du} c^r(q(u)) = (q_j^+ - q_j^-) D^r(q(u)) J_j^r(q(u)).$$

Integration gives the attribution column exactly as $\widetilde{C}^r_{:,j} = c^r(q^+) - c^r(q^-) = z_j^r$. Corollary 9 applied to their difference proves the stated maximum-error identity. No reference input derivative is needed to obtain this label.

For the local-to-joint implication set $A(q) = B[c^L(q) - c^S(q)]$. Equality of every separating boundary account of a sufficiently small single-coordinate change implies $\mathcal{K}_j(c^L(q+\epsilon e_j) - c^L(q) - c^S(q+\epsilon e_j) + c^S(q)) = 0$. The seminorm's kernel is $\ker B$: its expression forces all internal entries of Bh to vanish and its two endpoint entries to agree; their zero total then forces both to vanish. Hence $A(q + \epsilon e_j) = A(q)$. Differentiability and division by ϵ yield

$$\partial_j A(q) = B[D^L(q) J_j^L(q) - D^S(q) J_j^S(q)] = 0.$$

On any valid joint attribution path, integration of this identity gives zero divergence of the corresponding attribution-column error, and therefore matching accounts on every separating boundary. This needs the local matching premise throughout the domain containing the path; a finite training sample supplies no such premise.

B. Experimental methods

B.1. Network, path sets, and demand settings

The public Sioux Falls network has 24 nodes and 76 directed arcs (Transportation Networks for Research Core Team, 2024). All arcs belong to one cycle atom, hence one compatibility class. The eight queries are the positive-demand OD pairs with the largest published demands, with ties broken by ascending origin then destination node ID. All happen to involve node 10. The queries are $(10, 16)$, $(16, 10)$, $(10, 11)$, $(10, 15)$, $(15, 10)$, $(10, 17)$, $(11, 10)$, $(17, 10)$, with published demands $q_{\text{base}} = (4400, 4400, 4000, 4000, 4000, 3900, 3900, 3900)$ veh/h. The attribution path changes these demands from $0.2q_{\text{base}}$ to $1.6q_{\text{base}}$. Node IDs in the traffic experiments are one-based. Flows retain the benchmark's approximate-hourly convention; one time unit is 0.01 h. A reported TSTT number is $\sum_e F_e t_e(F_e)$ with F_e in veh/h and t_e in these time units: multiplying by 0.01 gives the hourly vehicle-hour total, without extrapolating to a day. All absolute attribution and boundary-account numbers use the same convention.

For each OD we retain every loopless directed path if enumeration finishes within 200 paths, otherwise the ten shortest by free-flow cost (Yen's algorithm, lexicographic edge-sequence tie breaking); no OD in this network has a fully enumerated set under the threshold. All eight query ODs use ten paths. The link–path incidence Δ and the matrix A summing paths within each OD stay fixed, so $B\Delta = MA$ with the appropriate full or study OD incidence matrix. The full-background ten-path set has 5,280 paths (ten per OD; the expanded set 10,560, twenty per OD); Appendix C.4 compares these candidate-path models.

The path-exchange response space $T_{\text{swap}} = \Delta \ker A$ (Appendix A.3) has dimension 32 in A with ten paths for each study OD, and dimension 53 in C with either ten or twenty paths for every OD, equal to the full circulation-space dimension $m - n + 1 = 53$ (ranks computed from within-OD differences of link–path incidence columns). Full background therefore spans the complete circulation space through allowable path exchanges, while the eight-OD path model spans a proper subspace.

The three demand settings share the same query endpoints. In setting A, all other demands are zero. In setting B, assign all positive background demands jointly with the study demands fixed at $q^0 = 0.2q_{\text{base}} = (880, 880, 800, 800, 800, 780, 780, 780)$ veh/h in the listed query order, freeze the resulting background link flows, and re-equilibrate only the eight query ODs. In setting C, the other 520 positive OD demands stay at published values while all 528 ODs re-equilibrate jointly at every path point and perturbation. TSTT and its gradient use total link flows in B and C; setting C is a full-background assignment with eight attribution queries, not an attribution evaluation of all 528 ODs.

B.2. Equilibrium and its sensitivities

Unless stated otherwise, the fixed-path logit stochastic user equilibrium (SUE) uses $\theta = 1$ and BPR costs $t_e(F_e) = t_{0,e}[1 + 0.15(F_e/c_e)^4]$. With Π_j the path set,

$$h_p = q_j p_p(F), \quad p_p(F) = \frac{\exp[-\theta(\Delta^\top t(F))_p]}{\sum_{r\in\Pi_j} \exp[-\theta(\Delta^\top t(F))_r]}, \quad F = \Delta h. \tag{B.1}$$

This is the solution of

$$\min_{h\geq 0,\ Ah=q} \left\{ \sum_e \int_0^{(\Delta h)_e} t_e(s)ds + \theta^{-1} \sum_p h_p(\log h_p - 1) \right\}.$$

For positive demands the feasible set is nonempty and compact, the objective strictly convex, and the minimizer positive: transferring mass to a zero path flow from a positive path of the same OD improves the objective, with entropy change $\theta^{-1}\epsilon \log \epsilon + O(\epsilon) < 0$. Its Hessian and sensitivity equations are

$$H = \Delta^\top \operatorname{diag}(t_e')\Delta + \theta^{-1} \operatorname{diag}(1/h_p) \succ 0, \qquad \begin{bmatrix} H & A^\top \\ A & 0 \end{bmatrix} \begin{bmatrix} \partial_j h \\ \partial_j \nu \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbf{e}_j \end{bmatrix}. \tag{B.2}$$

A has full row rank, so the system is nonsingular and the implicit function theorem gives a C^1 demand-to-flow map.

For background re-routing we solve the equivalent total-link-flow fixed point $F = L(F, q) = \Delta\{q_j p_{\Pi_j}(F)\}_j$ by Newton steps with residual line search. For frozen background the corresponding equation is $F = F^0_{\mathrm{bg}} + \Delta_{\mathrm{study}}\{q_j p_{\Pi_j}(F)\}_{j\in\mathrm{study}}$, where F^0_{bg} is fixed at the joint q^0 equilibrium. Costs and gradients use this total F in both equations. Every solution is checked against its path-loading equation, with maximum absolute path-flow gap below 10^{-7} veh/h. The 76-dimensional implicit system avoids forming a dense path Hessian. For an OD group R, define

$$S_R = \theta \sum_{k\in R} \Delta_k \left[\mathrm{diag}(h_k) - h_k h_k^\top / q_k\right] \Delta_k^\top, \quad W = \mathrm{diag}(t_e'(F_e)), \quad S = S_{\mathrm{all}}.$$

Differentiating the logit probabilities gives $\partial_F L_R = -S_R W$. Consequently the total sensitivity satisfies

$$(I + SW)J_j = \Delta_j p_j, \qquad J_j^{\mathrm{study}} = \Delta_j p_j - S_{\mathrm{study}} W J_j, \qquad J_j^{\mathrm{bg}} = -S_{\mathrm{bg}} W J_j. \tag{B.3}$$

Here "study" includes all eight queried ODs and "background" the other 520: $J_j = J_j^{\mathrm{study}} + J_j^{\mathrm{bg}}$, $BJ_j^{\mathrm{study}} = \mu_j$, $BJ_j^{\mathrm{bg}} = 0$, the study term including responses of the other seven study ODs. The decomposition uses one joint equilibrium, rather than sensitivities at different congestion states. For setting B the background flows have no derivative, so the same implicit system uses $S = S_{\mathrm{study}}$, with $J_j^{\mathrm{study}} = J_j$ and $J_j^{\mathrm{bg}} = 0$.

Central finite differences independently re-solve Eq. (B.1) at $q \pm a_j \mathbf{e}_j$, $a_j = 0.005 q_{\mathrm{base},j}$ and $a_j/2$; both steps are checked at every $\theta = 1$ ten-path re-routing integration point, the path-set and θ variations add half-step checks for all eight columns at the path midpoint, and the link Newton solver is compared against the successive-averages background equilibria at their common eleven quadrature points.

B.3. Conservative learned surrogate

The zero-background surrogate parameterizes $h_j = q_j \,\mathrm{softmax}(z(q)|_{\Pi_j})$, $f = \Delta h$. The three hidden layers each have 128 tanh units; the final layer is linear. The logit head uses training-standardized demands while the raw q_j multiplies the shares, preserving nonnegative flow and exact conservation. Computations use double precision. Training targets are fixed-path logit-SUE assignments ($\theta = 1$, path fixed-point tolerance 10^{-6}). The 2500/857/857 training/validation/test split uses seed 20260820, with demands $q_j = q_{\mathrm{base},j} s_j$ and independent s_j drawn from $\mathrm{LogNormal}(0, 0.5^2)$ clipped to $[0.2, 2]$. Adam uses learning rate 0.003, full-batch link-flow MSE, at most 12,000 steps, and validation every 50 steps. The minimum-validation-error state of each seed $0, \ldots, 4$ supplies the reported results, with automatic differentiation supplying their Jacobians; the test split is used only for final reporting.

B.4. Integration grids, connected boundaries, and cancellation diagnostics

All numerical integrations use composite trapezoids. The principal learned zero-background results use 201 quadrature points; direct zero-background SUE and both background settings use 41 in the main table. Frozen background compares 11 with 41 points; re-routing compares 11, 21, and 41 nested points; the two near-balanced queries add 81 and 161 nested points; the learned-versus-SUE sensitivity comparison uses identical 41-point grids, kept separate from the 201-point learned results (Appendix C.1).

Connected boundaries require both $G[U]$ and $G[V \setminus U]$ to be connected on the underlying undirected graph, with $s_j \in U$, $t_j \notin U$. Exact enumeration yields 10,270–13,253 admissible sets for each of the eight main study ODs, and 4,621–16,943 for the six additional queries of Appendix B.5. Extrema of $-\mathbf{1}_U^\top B\widetilde{C}_j$ are evaluated by exact enumeration, cross-checked against independent subset enumeration on restricted neighborhoods; attaining sets are verified for membership with values recomputed by direct summation. These are connected regions, not claimed to be administrative boundaries.

For cancellation diagnostics define $A_j = |\Delta q_j| \int_0^1 \sum_{v\notin\{s_j,t_j\}} |(BD_\lambda J_j)_v| d\lambda$ and $\kappa_j = 1 - S_j/A_j$ when $A_j > 0$. The triangle inequality gives $0 \le S_j \le A_j$. Both quantities use the specified 201-point learned grid and are summarized over the five seeds; checks on additional path-set or dispersion models sample the demand-path midpoint and endpoints.

Table B.1
Same-task single-OD computation. Relative errors use Euclidean norms; account error is maximum absolute geometric-account discrepancy in Sioux Falls units.

Method	Time (s)	Edge error	Node error	Account error
Endpoints	0.088	—	—	—
21-state integral	0.510	2.5e-08	1.7e-08	6.6e-05
41-state integral	0.948	6.3e-09	4.3e-09	1.6e-05
81-state integral	1.813	1.6e-09	1.1e-09	4.1e-06

B.5. Six pre-specified queries away from node 10

The selection rule uses only published demands and free-flow shortest-path times. Candidates are the positive-demand ODs whose origin and destination both differ from node 10; ranking them by demand and splitting into three equal-count terciles, the minimum- and maximum-free-flow-shortest-path OD within each tercile is selected, with ties broken by ascending (origin, destination). The six queries are $(7, 18)$, $(1, 19)$, $(4, 5)$, $(1, 15)$, $(6, 8)$, and $(7, 12)$, with published demands $(200, 300, 500, 500, 800, 700)$ veh/h and free-flow shortest-path times $(2, 22, 2, 23, 2, 19)$ time units; the joint protocol, sensitivities, and boundary enumeration are as in Section 6.2, with conservation checks at setting-C floating-point levels (maximum path fixed-point residual 1.6×10^{-10} veh/h, midpoint implicit-versus-finite-difference column error 2.6×10^{-8}, completeness 4.3×10^{-7}).

For the single-OD checks, let $\bar{q} = q^{\text{base}}$ be the full published demand vector over all 528 positive-demand ODs. For each focal query $j \in \{(1, 19), (1, 15), (7, 12), (6, 8)\}$, define

$$q^{(a)} = \bar{q} + (aq_j^{\text{base}} - \bar{q}_j)e_j, \qquad q^{(b)} = \bar{q} + (bq_j^{\text{base}} - \bar{q}_j)e_j, \qquad (a, b) \in \{(0.2, 1.6), (0.9, 1.1), (0.8, 1.2)\}, \quad \text{(B.4)}$$

where e_j is the coordinate unit vector. All other 527 demands, including the other five selected queries, remain at their published values $\bar{q}_{-j}$ along the segment. Direct equilibria and the learned small-band checks use these same reference states. The setting-C solver, path sets, and boundary definitions are unchanged; checks and interval results are in Appendix C.5. For each interval, Eq. (9) gives the attribution column from the two endpoint equilibria alone.

B.6. Same-task computation timings

Timings here describe the same fixed tasks and the same computational environment that produced the reported accounts (Section 5).

Endpoint versus sensitivity integration. For the single Sioux Falls query $(1, 19)$, only its demand changes from $0.8q_j$ to $1.2q_j$, with endogenous rerouting of the fixed published demands. Both methods share the path set, solver tolerance and 152-region geometric family; neither uses cached equilibrium states, each integral chain warm-starting only from its own preceding state, and each timed computation includes equilibrium solving, attribution formation, node multiplication and all geometric accounts. Common construction, excluded from both times, took 3.53 seconds for the fixed paths and 0.13 seconds for the region family. Timings are medians of three uncached runs; errors below compare sensitivity integration with the two-state endpoint attribution. The endpoint identity eliminates sensitivity integration for this separable single-OD task. It does not separate OD columns when several demands change jointly. The reported timings compare the methods on the same fixed task and computational environment.

Screenline aggregation. Table B.2 compares three aggregation methods on the 1642 oriented Sioux Falls regions of Section 5.3, once reference columns are available: the prefix scan; a vectorized baseline that precomputes the signed crossing matrix $A_{\mathcal{F}}$ (one sparse row per oriented region, $+1$ leaving and -1 entering) and aggregates all accounts with one sparse product $A_{\mathcal{F}}C$; and direct crossing-edge summation. All three agree to 2.2×10^{-11} TSTT units. The one-time matrix build (69.5 ms) serves only the vectorized method; the common candidate construction, projection sorting and connectivity preprocessing (83.48 ms) is required by all. Cold totals for six columns are 87.64 ms (scan), 153.22 ms (vectorized product including preprocessing and matrix construction) and 111.99 ms (direct); the table reports subsequent aggregation with the matrix

Table B.2
Same-task spatial aggregation after reference columns are available. Median batch times; reference generation is excluded. The 120-column workload repeats the six columns twenty times and measures repeated-query cost, not additional empirical evidence. Vectorized aggregates by one sparse product $A_{\mathcal{F}}C$ after a one-time 69.5 ms matrix construction.

Columns	Accounts	Prefix scan (ms)	Vectorized $A_{\mathcal{F}}C$ (ms)	Direct crossings (ms)
6	1642	4.16	0.24	28.51
120	32840	87.92	3.46	1176.15

already built. Among aggregations of a fixed region family, the precomputed matrix is fastest once repeated queries amortize its construction (at 120 columns, matrix construction plus aggregation costs about 72.9 ms against 87.92 ms for scanning; the six-column cold total does not repay it), the prefix scan needs no per-family matrix, and rebuilding each region's crossing edges inside the loop is slowest.

Anaheim aggregation. On the same 910 query–region accounts, with attribution already available, prefix scanning takes 0.10 ms, one precomputed sparse crossing-matrix product 0.30 ms, and direct crossing-edge aggregation 13.31 ms; the vectorized product agrees with node sums to 5.9e-12 and scanning with direct summation to 1.4e-12; building the crossing matrix takes 23.3 ms and the common distance-ordering and connectivity preprocessing 33.89 ms, required by all three algorithms. The original 21-state reference construction, including paths and sensitivities, took 1278.7 seconds; the additional refinement reused those states.

B.7. Anaheim network, equilibrium and predefined regions

Both network and demand files are the public Transportation Networks for Research Core Team Anaheim TNTP data (Transportation Networks for Research Core Team, 2024). The network header supplies free-flow times in minutes and capacities in vehicles per hour; all roads use the published BPR coefficient 0.15 and power 4. We use $t_e(F_e) = t_e^0[1 + 0.15(F_e/c_e)^4]$ and TSTT marginal weight $t_e^0[1 + 0.75(F_e/c_e)^4]$. The six study OD demands, ranked by descending base demand and then lexicographic OD, are $(4, 2)$: 2106.7, $(1, 2)$: 1365.9, $(2, 4)$: 1271.4, $(3, 2)$: 1237.9, $(25, 4)$: 1232.4, $(25, 2)$: 1232.3 vehicles per hour. They move jointly from 0.8 to 1.2 times base demand; all remaining positive demands stay at their published base values and retain endogenous route choice. Every OD uses ten loopless free-flow shortest routes from Yen's algorithm. Ties within 10^{-12} minutes are resolved by lexicographic edge-index sequence in the TNTP road order. Nodes 1–38 are centroids and cannot be intermediate route nodes; the first through node is 39. Logit scale $\theta = 1$ minute^{-1} applies to route travel time in minutes. The edge Newton solver targets maximum loading residual below 10^{-8} vehicles per hour, uses at most 80 iterations and backtracking up to 30 halvings per step, and checks the original path loading residual below 10^{-7}. Implicit derivatives solve the same fixed-point equation. Trapezoidal integration uses equally spaced states, with warm starts from a neighboring saved state. The 41-state refinement reuses all 21 old states and computes only the twenty new midpoints. The distance-based region construction and fixed progress targets are specified in Section 6.3. The reported range uses every admissible prefix in that family. Same-state mechanisms compare regions chosen nearest the progress targets, using free-flow weights t^0 and congestion weights equal to the total-flow TSTT marginal weight minus t^0.

B.8. Learning objectives and exact endpoint targets

Equation (20) suggests learning the reference's weighted demand response where it affects spatial accounts. With training-only positive scales a_j, the boundary term is

$$\mathcal{L} = \mathcal{L}_{\text{flow}} + \lambda_J \mathcal{L}_{\text{response}} + \lambda_{\text{cut}} \mathcal{L}_{\text{boundary}}, \qquad \mathcal{L}_{\text{boundary}} = \frac{1}{NK} \sum_{q,j} \left[\frac{\mathcal{K}_j(h_j(q))}{a_j} \right]^2 . \tag{B.5}$$

The reference Jacobian is obtained from the existing implicit SUE sensitivity system (Appendix B.2). We compare ordinary flow supervision, flow plus response supervision, and the full objective. Derivative supervision itself is established in Sobolev training (Czarnecki et al., 2017), including finite-difference derivative targets (Yuan et al., 2022); the network-derived target here selects the part of weighted response

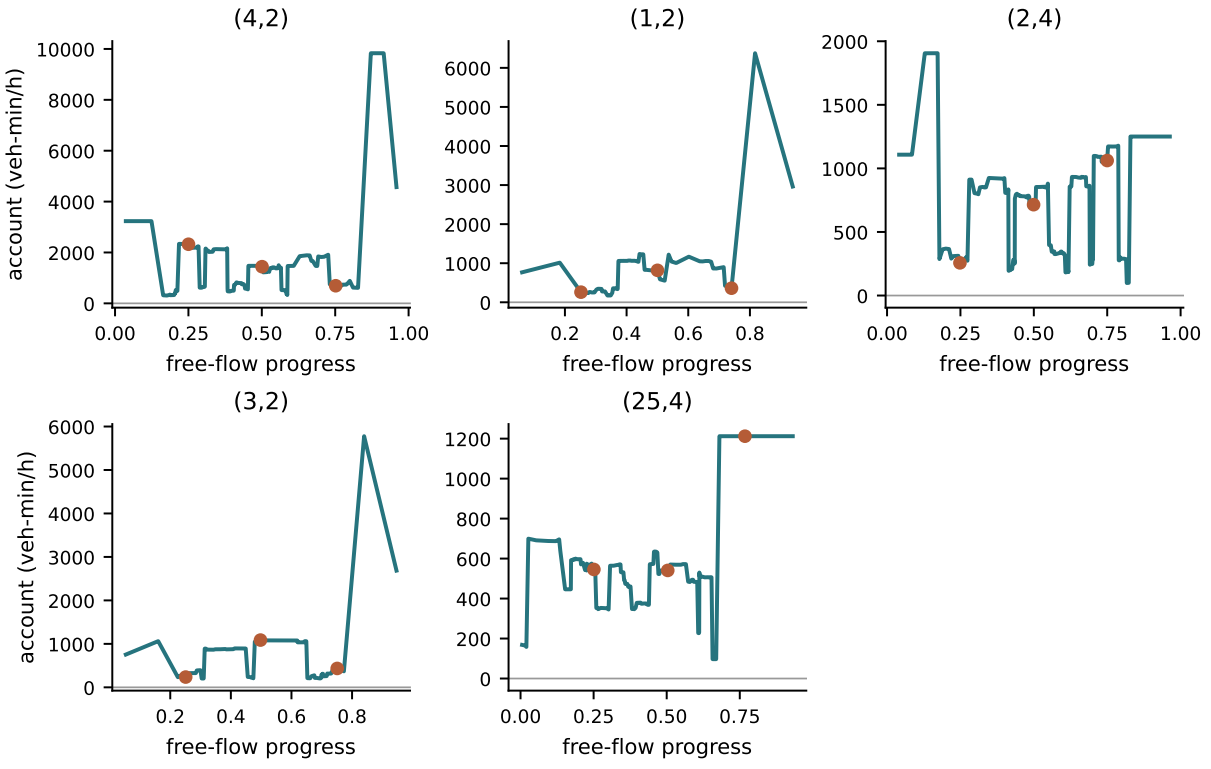


Figure B.1: Anaheim spatial response profiles of the remaining five study queries over the distance-based families: the account versus free-flow progress along each query's own OD direction, with ochre dots marking the three prespecified positions. All stay positive; family ranges span roughly one to one and a half orders of magnitude.

error that changes boundary accounts. A circulation in $\ker B$ is invisible to $\mathcal{K}_j$, so flow and response losses remain necessary for the other attribution objectives. The target matches the reference's actual response; it does not force its true internal attribution divergence to zero or replace TSTT weights with uniform weights. Equation (20) is a path inequality; finite training-point losses alone do not certify an entire demand domain.

For the following finite-change statement, the outcome is explicitly separable:

$$g(F) = (g_e(F_e))_{e\in E}, \qquad T(F) = \sum_{e\in E} g_e(F_e).$$

Proposition B.1 (exact finite-change boundary target)**.** *Let f^L, f^S and the edgewise outcome g be continuously differentiable on an open demand domain, and let $q^{\pm} = q \pm he_j$ with their entire segment in that domain. Put*

$$z_j^r = g(f^r(q^+)) - g(f^r(q^-)), \qquad r \in \{L, S\}.$$

For the attribution of this single-OD change,

$$\max_{U: s_j\in U,\ t_j\notin U} |\phi_j^L(U) - \phi_j^S(U)| = \mathcal{K}_j(z_j^L - z_j^S). \tag{B.6}$$

If all sufficiently small single-coordinate changes have matching accounts on every separating boundary at every point of the domain, then $BD^L J_j^L = BD^S J_j^S$ throughout the domain. Integrating yields matching boundary attribution columns for any joint demand path in that domain (with the same columnwise path convention).

Appendix A.7 proves both statements. The first is an exact finite-change target, whereas the pointwise response objective optimizes the upper bound in Eq. (20); integrating before taking $\mathcal{K}_j$ can retain

cancellations. The second statement requires a domain-wide local premise and is not a guarantee from finite training data. With several ODs changing, the endpoint outcome difference only identifies the sum of their attribution columns.

For sampled single-OD endpoint pairs, a direct training objective is

$$\mathcal{L}_{\text{pair}} = \mathcal{L}_{\text{flow}} + \lambda_{\text{edge}} \frac{1}{N} \sum_{(q,h,j)} \frac{\|z_j^L - z_j^S\|_2^2}{b_j^2} + \lambda_{\text{cut}} \frac{1}{N} \sum_{(q,h,j)} \left[\frac{\mathcal{K}_j(z_j^L - z_j^S)}{a_j} \right]^2, \tag{B.7}$$

with positive training-only scales a_j, b_j. The edge term controls changes invisible to the boundary seminorm, whose kernel is $\ker B$. Two reference endpoint equilibria supply z_j^S; training uses ordinary parameter backpropagation through two predictions, without reference Jacobian labels or input-Jacobian differentiation. Actual label-generation and training costs must still be measured.

B.9. Learned-model evaluation protocol

Four objectives continue each of five common conservative models for 1,200 Adam updates, with learning rate 3×10^{-5}, flow batch 32 and query batch eight. The flow data and seed-specific minibatch sequences are shared. Each of six queries has 64 training and 16 validation endpoint pairs. Centers are sampled coordinatewise from $[0.65, 1.3]$ times baseline demand (seed 102111); half-widths 2.5% and 15% occur equally often. Only one demand changes per pair, while background routes remain responsive. Endpoint-edge and endpoint-edge-plus-boundary objectives use identical endpoint labels, coefficients one and 0.01, and training-only root-mean-square scales. Response supervision uses derivatives at the same pair centers; its coefficient is one.

The flow loss averages $[(f_e^L - F_e^S)/s_e]^2$ over minibatch states and edges, with $s_e = 0.01\sqrt{\text{mean}_{\text{train}}[(F_e^S)^2]}$ from the 2,500 training states. The response loss is the batch mean of $\|J_j^L - J_j^S\|_2^2/u_j^2$. With $\langle\cdot\rangle_j$ denoting the average over query j's training pair centers or endpoint labels, the scales are

$$u_j = \max\{\sqrt{\langle\|J_j^S\|_2^2\rangle_j}, 10^{-6}\},$$
$$b_j = \max\{\sqrt{\langle\|z_j^S\|_2^2\rangle_j}, 10^{-6}\}, \qquad a_j = \max\{\sqrt{\langle\mathcal{K}_j(z_j^S)^2\rangle_j}, 10^{-6}\}.$$

Edge and response losses sum squared errors over edges before batch averaging; their coefficients are one, while the boundary coefficient is 0.01.

Candidates comprise the common initial model (update zero) and every 100th continuation update. We minimize normalized validation boundary mean-square error among candidates with mean endpoint flow error at most 1.1 times the initial model's value, using the same 96 validation pairs for every method. The initial model ensures this set is nonempty. Test paths never enter selection.

The 24 fixed test paths (seed 102112) contain six each with one, two, three and six active queries; cyclic combinations balance query participation. Changes are 15–30% of baseline demand with both same-direction and mixed-direction paths, each with 41 states. For every path, flow error first averages relative Euclidean errors across its states. Attribution statistics average active queries and seeds within each path and then take medians across the 24 paths; seeds do not add independent paths. For column error e_j and reference column c_j, relative column, system and unrestricted-boundary errors divide $\|e_j\|_2$, $|\mathbf{1}^\top e_j|$ and $\mathcal{K}_j(e_j)$ by the corresponding reference values, with absolute floor 10^{-6}. Geometric error uses $\max_{U \in \mathcal{F}_j^{\text{geo}}} |\mathbf{1}_U^\top B e_j|$ and the same maximum for c_j. Percentages multiply ratios by 100.

The decomposition in Eq. (C.1) is evaluated at the same 41 states with the same quadrature. Component norms describe error mechanisms; cancellation prevents treating their norms as additive shares. The fixed-screenline sign comparison in Appendix C.7 uses the separate frozen six-query baseline segment and is not pooled with these 24 paths. Selected updates in seed order 0–4 are Flow: 1100, 800, 1000, 100, 700; Flow + endpoint edge: 0, 1200, 0, 0, 1100; Flow + endpoint edge + boundary: 0, 1200, 0, 0, 1100; Flow + response: 1100, 1200, 1200, 1200, 1200. All selected models meet the endpoint flow constraint. For each endpoint objective, three seeds retain update zero. On the original held-out demand-state split, seed-mean relative flow errors are Flow: 0.0386%, Flow + endpoint edge: 0.0445%, Flow + endpoint edge + boundary: 0.0445%, Flow + response: 0.0388%. These are distinct from the matched-path flow errors in the table. Training labels

Table B.3
The paper's principal learning comparison: four objectives with update zero included in validation selection. All prediction and explanation entries use the same 24 paths and report path medians after averaging seeds and active queries.

Method	Flow (%)	Column (%)	System (%)	Boundary (%)	Geometric (%)
Flow	0.0325	21.61	2.33	14.53	7.91
Flow + endpoint edge	0.0376	21.80	2.92	18.91	8.30
Flow + endpoint edge + boundary	0.0376	21.81	2.95	18.93	8.30
Flow + response	0.0291	14.22	3.31	12.08	5.81

Table B.4
Incremental computation costs (seconds). Training-label and common-validation costs are measured separately on the same CPU protocol; training times are archived continuation medians, and explanation times are newly measured sequential medians for the selected models.

Method	Training labels	Common validation	Continuation	Explanation
Flow	0.00	6.45	246.0	7.34
Flow + endpoint edge	25.75	6.45	422.2	7.59
Flow + endpoint edge + boundary	25.75	6.45	275.1	7.27
Flow + response	16.12	6.45	297.1	7.13

comprise 768 endpoint states for either endpoint objective, or 384 center states and their sensitivities for response supervision. Every method additionally uses the same 192 endpoint states for validation; unused center-validation derivatives are excluded. Endpoint objectives share their labels and do not incur duplicate reference solves. Existing flow data, pretrained models and the common test reference are excluded from these incremental costs. The 984 test states and sensitivities took 74.4 s in the archived reference run. Explanation timing covers all 24 paths, forward evaluation, six directional derivatives per state, quadrature and node aggregation, excluding model loading. Archived continuation times include initialization, validation and saving under concurrent seed jobs; they describe the actual training expense and do not establish a method speed advantage.

Across the five selected seeds, the ranges of mean unrestricted-boundary errors are $[13.36, 24.12]\%$, $[12.29, 30.50]\%$, $[12.27, 30.50]\%$, $[10.53, 17.46]\%$, respectively, in the method order of Table B.3. Each seed statistic first averages active queries within each path and then gives equal weight to the 24 paths. These ranges describe seed variability rather than additional independent test paths.

C. Numerical accuracy and supporting results

C.1. Grid refinement, derivative, and solver checks

All differences here are observed numerical comparisons, not certified bounds. Direct attribution-column differences are at most 0.178% for 11 versus 41 states and $1.1 \times 10^{-4}\%$ for six-query/single-OD 41 versus 81 states. Near-zero accounts on 41/81/161 grids change by less than 5×10^{-6} percentage points (absolute < 0.01); the learned 201/801-state relative difference is 4.97×10^{-6}. Implicit versus re-solved central-difference Jacobian errors are at most 7.6×10^{-5} (halved step 2.5×10^{-5}), or 3.0×10^{-8} for six-query/single-OD columns. Newton versus successive-averages differences over 1,353 solves are at most 1.31×10^{-11}. Maximum fixed-point and balance residuals are 3.10×10^{-8} and 1.82×10^{-11} veh/h. Scalar completeness relative discrepancies are at most 3.36×10^{-5} for frozen/re-routing settings and 1.9×10^{-7} for single-OD checks. Per-edge endpoint checks give relative column errors 7.7×10^{-8}–1.1×10^{-6} and maximum entry residuals 5.4×10^{-4}–1.6×10^{-2} TSTT units. Per-query records remain in the arrays. The direct $(10, 16)$ account change is 24.80% on its 41-state grid; Figure C.1 uses the seed-zero learned model on 201 states (24.8%).

C.2. Normalization, cancellation, and sensitivity

Complete cut-range table. Table C.1 completes the three representative rows of Table 9 with all eight study ODs.

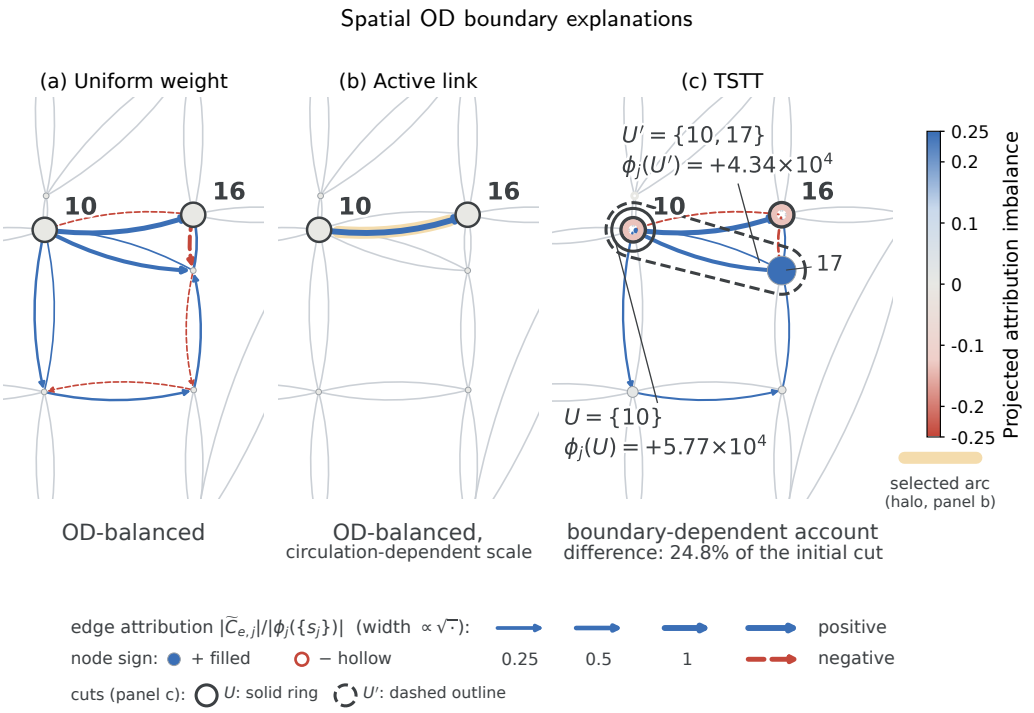


Figure C.1: OD $(10, 16)$ endpoint attributions, zero background, learned seed 0. (a) Uniform weights give cut consistency. (b) The active-link outcome balances in the OD direction but has no response-independent scale. (c) The signed TSTT boundary account decreases by 24.8% when the boundary expands from $\{10\}$ to $\{10, 17\}$. Widths encode attribution magnitude; node colors show the projected attribution imbalance $(d_j)_v$ (Eq. (12)) divided by the absolute origin-cut value.

Table C.1
TSTT cut ranges as percentages of each account's absolute origin-cut value, all eight study ODs (complete version of Table 9). A: zero background; B: frozen background flows; C: background demands fixed, all traffic re-equilibrated. Learned values are five-seed medians (ranges); all SUE columns are direct (41 quadrature points, implicit sensitivities).

	Unrestricted range (%)				Connected (%)
OD	Learned A	SUE A	SUE B	SUE C	SUE C
$(17, 10)$	59.71 (58.92–61.23)	59.95	56.38	54.77	35.27
$(10, 17)$	37.55 (37.40–37.94)	37.48	43.62	49.11	31.57
$(16, 10)$	36.80 (36.41–38.40)	36.86	39.99	166.26	127.72
$(10, 16)$	26.85 (26.68–27.07)	26.94	34.02	109.04	83.80
$(15, 10)$	0.84 (0.69–1.16)	0.77	21.98	84.05	58.24
$(10, 15)$	0.81 (0.76–0.88)	0.77	18.89	71.88	49.90
$(10, 11)$	0.23 (0.10–0.35)	0.0075	1.17	84.67	51.27
$(11, 10)$	0.18 (0.03–0.27)	0.0073	0.79	81.09	48.98

Error-metric definitions. The surrogate test flow error is, within each seed, the mean over the 857 test demands of $\|f(q) - F^{\mathrm{SUE}}(q)\|_2 / \|F^{\mathrm{SUE}}(q)\|_2$, reported as the median over the five seeds; Jacobian column errors $\|J_j^{\mathrm{learned}} - J_j^{\mathrm{SUE}}\|_2 / \|J_j^{\mathrm{SUE}}\|_2$ are pooled over the $5\times41\times8$ seed–point–OD columns; attribution-column errors use the per-column norm over the 40 seed–OD columns on the common 41-point grid.

Same-path flow, sensitivity, and boundary-account errors. The test-split flow error is not paired with the Jacobian comparison; re-summarized on the same 41 zero-background quadrature points for each seed (median / 95th percentile / maximum, percent): flow error 0.136 / 0.682 / 1.331; Jacobian column error 0.516 / 3.903 / 9.562 (maximum at seed 3, OD $(16, 10)$, $\lambda = 0.925$; $5 \times 41 \times 8$ seed–point–OD units); attribution-column errors median 0.349%, maximum 1.80%; and E_j^{cut}, which evaluates both models' accounts at every connected boundary of a query's family on the same path (the unrestricted-family counterpart is virtually identical), at 0.204 / 0.650 / 1.739 (40 seed–OD units; largest connected error at OD $(16, 10)$, seed 3).

Table C.2 supplies the per-query absolute origin-cut denominators for Table 9. The setting-C link sums C_j span 2.078–4.282 $\times 10^5$ TSTT units, exceeding origin-cut accounts by factors of 1.01–2.37; their sum matches the endpoint TSTT change within 2.3×10^{-5} relative.

Table C.2
Absolute direct-SUE origin-cut accounts $|\phi_j(\{s_j\})|$, in TSTT units, for the eight queries in Table 9. A, B, C use their respective 41-point columns; entries rounded to one decimal.

OD	A	B	C
$(17, 10)$	29,372.1	249,729.2	224,954.3
$(10, 17)$	46,977.2	317,861.9	245,178.7
$(16, 10)$	41,994.3	352,858.5	180,671.5
$(10, 16)$	57,468.9	405,374.2	272,743.2
$(15, 10)$	33,859.4	226,878.2	201,351.2
$(10, 15)$	34,056.4	258,682.3	232,160.6
$(10, 11)$	28,807.1	339,651.4	202,928.2
$(11, 10)$	28,010.4	324,427.0	204,802.3

The node breakdown in Table C.3 complements the fixed boundary comparison of Section 6.2.

Table C.3
Signed node divergence in the re-routing setting, at the same joint equilibrium state and total-flow gradient at each of the 41 quadrature points (study includes all eight query ODs; TSTT units of Appendix B.1; entries rounded independently, so a component sum may differ from the rounded total in its last digit).

OD	Node	Study	Background	Total
$(10, 16)$	15	-973.1	$49,103.4$	$48,130.3$
$(10, 16)$	8	9.6	$42,102.7$	$42,112.3$
$(10, 16)$	20	0.2	$41,805.4$	$41,805.6$
$(10, 11)$	5	0.5	$35,952.1$	$35,952.6$
$(10, 11)$	12	0.0	$-23,858.1$	$-23,858.1$
$(10, 11)$	9	-0.2	$-18,557.6$	$-18,557.7$

For the learned zero-background low-range group, five-seed median pointwise imbalance budgets $100A_j/|\phi_j(\{s_j\})|$ lie between 0.22% and 0.95%, versus 26.85–59.71% for the high group; the low-group median cancellation factors are at most 0.313 (one seed reaches $\kappa = 0.852$), so small pointwise imbalance mainly limits the low-group range, while the high group's median κ is below 8×10^{-5}.

Zero-background route-choice dispersion checks at $\theta \in \{0.5, 1, 2\}$ (eleven-point direct-SUE calculations) retain the large/small separation — large ranges 27–64%, the others below 9.4%.

C.3. Full-background six-query surrogate validation

The six-query conservative surrogate predicts shares over all 5,280 paths, with the architecture of Appendix B.3, dataset seed 20261025 and batch size 256. Its 4,214 target solves have maximum path residual 3.5×10^{-8}; learned conservation residual is at most 3.6×10^{-11}. Held-out flow error is 0.040% (seed median; range 0.035–0.056%). On the 41 common path states, flow errors are 0.037–0.065%, but increment errors are 5.0–7.0% against a reference increment norm 2,982.7 (2.7% of flow level). Median column errors are 81.2% for $(7, 18)$, 31.1% for $(4, 5)$ and 7.4–11.8% for the others. System contribution errors reach 14.7% and 10.3% for those two queries, versus at most 3.7% for the sign-crossing queries. Table C.7 retains extrema and boundary errors; learned column completeness is within 6×10^{-6} relative.

An exact two-term split of (2) (L/S: learned/direct) identifies where this error arises:

$$\widetilde{C}^L_{:,j} - \widetilde{C}^S_{:,j} = \Delta q_j \int_0^1 (D^L_\lambda - D^S_\lambda) J^S_j \, \mathrm{d}\lambda + \Delta q_j \int_0^1 D^L_\lambda (J^L_j - J^S_j) \, \mathrm{d}\lambda, \tag{C.1}$$

satisfied entrywise within 10^{-10} TSTT units. The weight-shift term has querywise median norm at most 0.2% of the reference column; the response term is larger in every one of the thirty seed–query units. Their

norm sum is at most 1.04 times the realized error norm. Signed scalar terms can cancel, so these norms are not additive shares. Small-band endpoint columns have 4.8–15.9% error and system values within 2.7% of reference, while every seed retains the reported signs.

C.4. Sensitivity to candidate paths and route-choice dispersion

The path-set comparison expands all 528 OD sets to twenty candidates each (the first ten unchanged), with demands, BPR functions, $\theta = 1$, and 41 quadrature points fixed: across the eight queries, absolute unrestricted and connected ranges change by at most 0.977% and 1.011%, with the three largest internal-node terms keeping their ordered node IDs (moving at most 1.43%). The $\theta \in \{0.5, 2\}$ check splits the specified $(10, 16)$ boundary difference into study/background components $-7{,}404.9/-2{,}371.5$ at $\theta = 0.5$ and $+115.3/+235.7$ at $\theta = 2$; for all three additional models, implicit derivative columns match re-solved half-step central differences at the demand-path midpoint (maximum 0.00249%, completeness gap 2.85×10^{-5}).

C.5. Six-query interval, mechanism, boundary-family, and scenario checks

Extremal ranges and negative shares. Unrestricted extrema follow the closed form (14), connected extrema the same enumeration as the connected ranges (Table C.4); the negative shares in Section 6.2 are exact counts over the connected families (4,621–16,943 admissible sets per query), and all displayed values come from the 41-point columns.

Table C.4
Minimum–maximum boundary-account ranges for the three sign-crossing queries of Table 5, in TSTT units (Appendix B.1), rounded to integers. Baseline: unrestricted extrema (closed form (14); connected extrema in Table 5); next four rows: connected ranges under the model-setting changes and the single-OD $0.2 \to 1.6$ re-run (unrestricted intervals within 1% of the baseline row); last two: small-interval bands, with C_j positive (+7,202 to +33,734 across the three queries) and the control $(6, 8)$ positive ([5,079, 9,623] and [10,157, 19,246]; single-OD [35,352, 66,861]).

Range	$(1, 19)$	$(1, 15)$	$(7, 12)$
Baseline (10 paths, $\theta = 1$): unrestricted	[−19,116, 26,696]	[−16,240, 44,137]	[−45,241, 61,650]
Twenty paths, $\theta = 1$: connected	[−13,451, 24,276]	[−8,329, 40,644]	[−17,728, 59,452]
Ten paths, $\theta = 0.5$: connected	[−13,024, 24,064]	[−8,107, 39,933]	[−17,633, 60,476]
Ten paths, $\theta = 2$: connected	[−13,016, 24,411]	[−8,233, 40,892]	[−19,587, 61,647]
Single-OD $0.2 \to 1.6$: connected	[−13,608, 24,414]	[−8,359, 40,745]	[−17,580, 60,006]
Single-OD ±10% band ($0.9 \to 1.1$): connected	[−1,947, 3,494]	[−1,201, 5,838]	[−2,517, 8,608]
Single-OD ±20% band ($0.8 \to 1.2$): connected	[−3,895, 6,988]	[−2,402, 11,677]	[−5,035, 17,215]

Connected minimum and divergence decomposition. Table C.5 reports, for each sign-crossing query, the connected minimizing boundary U_- at the baseline setting: its account, and the interior divergence sum $\phi_j(\{s_j\}) - \phi_j(U_-)$, decomposed as the table caption specifies, at the same joint state and gradient as the reported columns (additivity $\leq 1.2 \times 10^{-10}$). By Eq. (15), the negative divergence captured inside U_- (4,664.9, 6,960.1, and 19,811.5 for the three queries) covers 83%, 88%, and 72% of the gaps $\phi_j(U_-) - \phi_j^{\min}$, while the captured positive divergence (19,938.0, 18,194.2, 46,901.4) is 95.5%, 94.8%, and 85.7% of the query totals. The sign change arises differently across queries: study response dominates for $(1, 19)$ (largest node 15: own +14,464.0), study- and background-dominated nodes share the sign for $(1, 15)$, and both contribute about equally for $(7, 12)$ (node 8). The extremal node sets (one-based node IDs) are $U_-(1, 19) = \{1, 2, 6, 7, 8, 9, 15, 16, 18, 20, 21, 22, 24\}$ and $U_+(1, 19) = \{1, 2, 3, 4, 5, 6, 10, 11, 12, 13, 14, 22, 23\}$; $U_-(1, 15) = \{1, 2, 6, 7, 8, 9, 16, 17, 18, 19, 20, 21, 24\}$ and $U_+(1, 15) = \{1, 2, 3, 4, 5, 6, 10, 11, 12, 13, 14, 22, 23\}$; $U_-(7, 12) = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 14, 17, 19, 23\}$ and $U_+(7, 12) = \{7, 8, 9, 15, 16, 18, 20, 21, 22, 24\}$.

Neighbourhood families. Use undirected hop distance or shortest free-flow distance from the origin (minimum weight over parallel arcs). Include nodes at each attained radius, stop before including the destination, and retain only two-sided connected regions. Table C.6 reports all family ranges; all are positive with minima at the origin cut. For short-path queries $(7, 18)$, $(4, 5)$ and $(6, 8)$, only that cut is admissible. The thirteen-node free-flow ball at radius 16 for $(1, 19)$ is $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 24\}$, with account +10,756.0.

Straight screenlines. Coordinates use an equirectangular frame (longitude scaled by the cosine of mean latitude). Section 5.3's 36-direction construction, with duplicate regions removed, gives 35, 152, 32, 123, 23

Table C.5
Connected minimizing boundary U_- at the baseline setting for the three sign-crossing queries, in TSTT units. Interior is the divergence sum over $U_- \setminus \{s_j\}$, decomposed into the query's own response, the other five query ODs, and the background response; the study group of the main text is the sum of the first two columns. Node sets are the one-based lists in the preceding paragraph. Entries are rounded independently.

OD	$\phi_j(\{s_j\})$	$\phi_j(U_-)$	Interior	Own OD	Other query	Background
$(1, 19)$	1,771.6	−13,501.6	15,273.1	19,480.9	−59.4	−4,148.3
$(1, 15)$	2,948.1	−8,286.0	11,234.1	7,856.9	−26.7	3,404.0
$(7, 12)$	9,473.8	−17,616.2	27,089.9	14,002.3	−3.2	13,090.8

Table C.6
Origin-centred neighbourhood accounts, baseline six-query setting C (TSTT units). Counts are distinct admissible node sets within each family; sets shared by the two families are counted in each row.

OD	Neighbourhood	Count	Minimum	Maximum
$(1, 19)$	Hop distance	6	$1,771.6$	$15,803.5$
$(1, 19)$	Free-flow time	13	$1,771.6$	$15,803.5$
$(1, 15)$	Hop distance	5	$2,948.1$	$26,602.0$
$(1, 15)$	Free-flow time	14	$2,948.1$	$26,602.0$
$(7, 12)$	Hop distance	4	$9,473.8$	$31,524.1$
$(7, 12)$	Free-flow time	9	$9,473.8$	$32,139.4$

and 148 regions in Table 5 order. All accounts agree with direct incidence aggregation below 10^{-9}. Refining directions from 5° to 1° leaves both negative minima unchanged; negative shares change from 1.3% to 1.8% for $(1, 19)$ and 6.8% to 6.2% for $(7, 12)$. Other queries stay positive; only $(7, 18)$'s minimum changes, from 642.4 to 641.4.

Model-setting checks. Re-solving the protocol of Appendix B.5 under twenty candidate paths per OD and $\theta \in \{0.5, 2\}$ at ten paths gives the ranges of Table C.4 (C_j within 3.9%); solver and finite-difference checks match Appendix B.5 levels.

Single-OD demand changes. Each single-OD demand path ($0.2 \to 1.6$ of published demand) re-solves the setting-C protocol with one moving coordinate. For the three crossing queries the connected ranges (Table C.4, single-OD row) still straddle zero, their extrema attained by exactly the joint-scenario sets U_- and U_+ (changes at most 0.9% and 0.5%; unrestricted intervals within 1% of the joint baseline) and the contributions stay positive (C_j = 50,322.2, 77,672.6, 117,476.8); the control $(6, 8)$ remains sign-consistent (C_j = 74,176.1, connected range in the caption). The small-interval bands (Table C.4, last two rows) follow the same protocol, the 41-point check ($(1, 19)$, $0.8 \to 1.2$) reproducing the endpoint column to 1.3×10^{-5} TSTT units per edge.

Endpoint identity checks. Appendix C.1 reports the per-edge endpoint-identity errors and their relation to completeness gaps on these same demand paths.

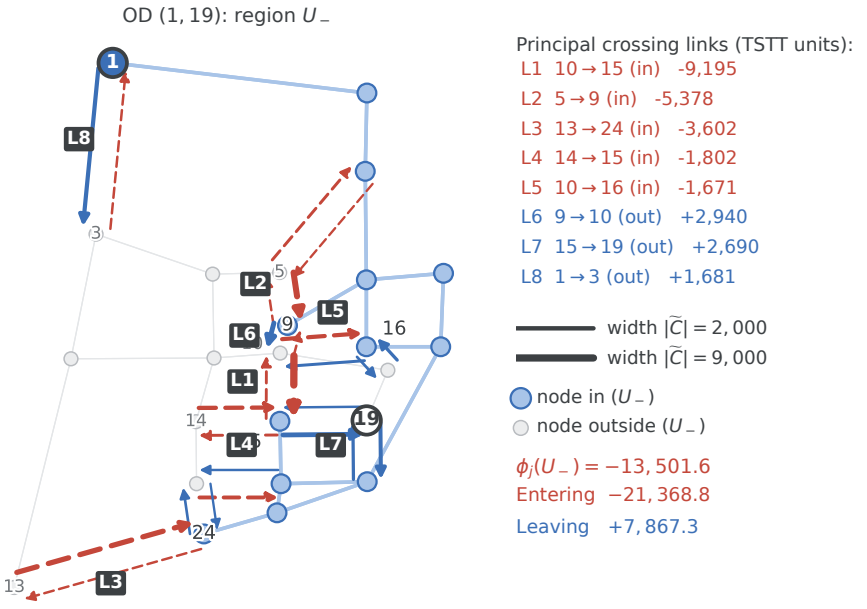


Figure C.2: The connected minimizing region U_- for OD $(1, 19)$ (setting C, six-query protocol). Shaded nodes form U_- (origin node 1 ringed; destination 19 outside). Crossing-link arrows keep the link direction; width scales with $|\widetilde{C}_{e,j}|$, and colour and style encode the signed contribution to $\phi_j(U_-)$, $a_{e,j} = \sigma_e \widetilde{C}_{e,j}$ ($\sigma_e = +1$ leaving, -1 entering; solid blue positive, dashed red negative), so a positive attribution on an entering link appears red. The side panel lists, in TSTT units, the five largest negative and three largest positive contributions among the 26 crossing links (L1–L5 negative, L6–L8 positive, "(in)"/"(out)" marking entering and leaving links), with the width scale and the membership legend.

C.6. Baseline surrogate envelopes

Table C.7
Full-background surrogate validation on the identical 41-state path of the six-query protocol (setting C; TSTT units; contributions and extrema rounded to integers, errors to one decimal). C_j^{learned} and the connected extrema: across-seed envelopes of per-model values (direct-SUE values in Table 5). Last two columns: five-seed *median* of the largest connected-boundary disagreement ϵ_j^{cut}, in absolute units and as a percentage of the direct connected span.

		Connected extrema: across-seed ranges			
OD	C_j: across-seed envelope	per-model minima	per-model maxima	ϵ^{cut} (med.)	ϵ^{cut}/span (med., %)
$(7, 18)$	[574, 681]	[335, 521]	[907, 1,299]	397.5	522.5
$(1, 19)$	[49,929, 51,452]	[−13,911, −13,334]	[23,683, 23,948]	1,567.4	4.2
$(4, 5)$	[2,535, 2,934]	[1,654, 2,148]	[2,946, 3,546]	1,050.8	99.5
$(1, 15)$	[77,416, 80,233]	[−7,717, −6,766]	[39,473, 40,076]	2,187.3	4.5
$(6, 8)$	[73,565, 75,224]	[36,412, 37,631]	[66,892, 67,762]	2,163.2	6.9
$(7, 12)$	[116,562, 118,706]	[−17,084, −15,198]	[59,965, 61,169]	4,111.3	5.3

C.7. Fixed-screenline surrogate comparisons

Sign preservation. All five baseline seeds retain the three connected-family sign crossings and positive system contributions. At the fixed reference extrema, $m_j = \min\{-\phi_j^S(U_-), \phi_j^S(U_+)\} - \epsilon_j^{\text{conn}}$ is positive, with smallest value 5519 TSTT units. This verifies sign preservation without implying accurate column magnitudes.

At the fixed direct-model minimizing screenline, every baseline seed retains the negative sign for both queries: $(1, 19)$ gives $[-3886.6, -3365.1]$ against the reference -2766.6, and $(7, 12)$ gives $[-2810.4, -1916.6]$ against -2967.3. Their largest family-wide errors across seeds are 1430.8 and 2499.2, below the respective

Table C.8
Fixed straight screenlines: reference minimum and the five frozen flow-supervised surrogates at that same screenline (TSTT units). The last column gives the largest error over the screenline family, as an across-seed range.

OD	Reference minimum	Learned account: seed range	Maximum screenline error: seed range
$(7, 18)$	642.4	[576.9, 789.8]	[141.0, 269.4]
$(1, 19)$	-2766.6	[−3886.6, −3365.1]	[857.9, 1430.8]
$(4, 5)$	2652.3	[2301.7, 2918.8]	[438.0, 648.3]
$(1, 15)$	511.1	[732.8, 2068.2]	[1140.6, 2553.0]
$(6, 8)$	45230.2	[46431.5, 47218.3]	[1276.2, 1988.1]
$(7, 12)$	-2967.3	[−2810.4, −1916.6]	[1736.8, 2499.2]

reference sign margins. These are screenline-specific certificates; they supplement the separate connected-family result above (Appendix C.7).

The eight-query path-set and dispersion checks are collected in Appendix C.4. Boundary variation persists, although the specified $(10, 16)$ comparison $\{10\} \to \{10, 17\}$ changes sign across θ; these are different fixed-path models, not convergence to unrestricted SUE.

C.8. Additional learning comparisons

Two additional pointwise comparisons use five conservative models, 600 Adam updates at 3×10^{-5}, flow/response batches of 32/8 and training-only scales. Derivative labels comprise 256 training and 64 validation states (seed 20261029). The fixed-coefficient comparison uses coefficients one; calibration selects 0.01 from $\{0.001, 0.01, 0.1\}$ with seed zero under a 110% relative flow-error threshold, then applies it across seeds (seven boundary fits). Each comparison tests four prescribed 41-state paths, with one, two, three and six active queries and baselines in $[0.65, 1.3]$ times published demands (test seeds 101029/101030). Fixed-coefficient boundary supervision yields 22.25% boundary error versus 9.64% for response supervision; calibration yields 7.74% versus 7.44% on its new paths. Neither comparison supports overall improvement. Complete current four-method results, with common selection and matched-path metrics, are in Table B.3.

C.9. Complete input-stratified common-summary results

Positive non-diagonal published demands are stratified by demand and directed free-flow shortest-path time using empirical tertiles, with ties kept in the lower bin. Within each of nine cells, lexicographically sorted OD indices at zero-based positions $\lfloor N/3 \rfloor$ and $\lfloor 2N/3 \rfloor$ are retained. No attribution values enter selection. Endpoint differences identify total edge attribution, without identifying the same-state study/background response split. The two perturbation widths share every field of the selection protocol and differ only in the demand interval; their normalized errors are close for every query, so Table C.9 reports both in one table. Absolute minima, maxima, midpoints and half-ranges for both widths, and the per-position account sequences behind the segmentation of Proposition 10, accompany the result arrays in the code package.

Table C.9
Complete geometric common summaries for the 18 input-stratified queries at both widths. Normalized error is Eq. (22) at $\pm 10\%$ and $\pm 20\%$; ranges are $[\phi_{\min}, \phi_{\max}]$ at $\pm 20\%$ in Sioux Falls cost units; P and M mean positive and mixed signs at $\pm 20\%$.

OD	System	Range ($\pm 20\%$)	Norm. err. ($\pm 10\%$)	Norm. err. ($\pm 20\%$)	Sign
(4,3)	507.2	[429.0, 535.0]	9.9	9.9	P
(12,3)	337.2	[330.7, 340.2]	1.4	1.4	P
(6,1)	2360.6	[658.3, 2290.4]	35.6	35.6	P
(15,6)	9390.2	[415.4, 4433.5]	45.3	45.3	P
(6,21)	4577.8	[-3.4, 2349.4]	50.1	50.1	M
(18,14)	3422.3	[150.8, 1929.7]	46.1	46.1	P
(8,4)	25097.5	[1270.7, 15741.3]	46.0	46.0	P
(19,8)	16939.7	[733.2, 7266.4]	45.0	45.0	P
(11,6)	6165.9	[1236.4, 6106.4]	39.9	39.9	P
(16,2)	17763.1	[896.3, 9628.6]	45.3	45.3	P
(8,21)	7605.5	[752.6, 4214.1]	41.1	41.1	P
(14,8)	17445.3	[890.3, 7527.1]	44.1	44.1	P
(11,15)	47925.5	[17628.9, 27737.8]	18.2	18.2	P
(17,9)	24040.9	[5482.3, 18519.3]	35.2	35.2	P
(10,12)	55214.5	[9822.4, 33248.6]	35.2	35.2	P
(13,10)	54411.9	[2918.8, 31345.2]	45.3	45.3	P
(1,10)	33917.2	[2182.9, 20856.4]	44.8	44.8	P
(10,1)	34264.5	[2182.7, 21039.5]	44.8	44.8	P

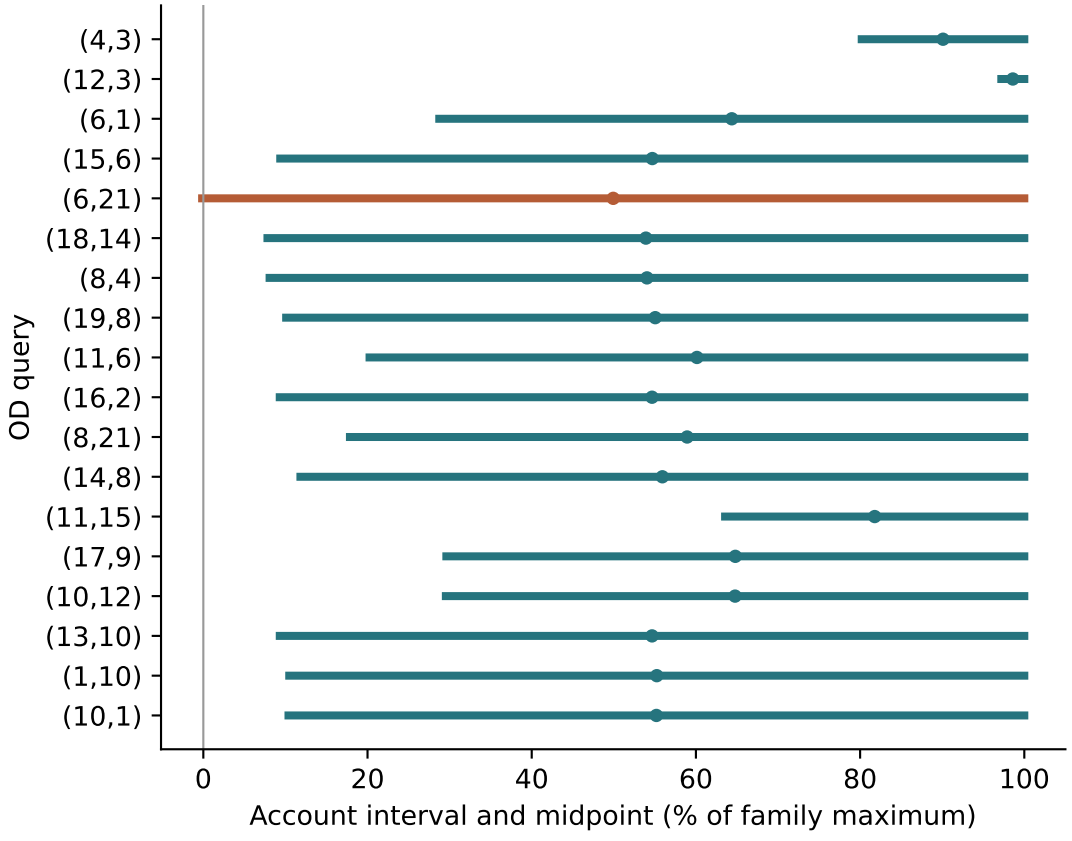


Figure C.3: Geometric account intervals normalized by each query's largest account magnitude, over all 18 input-stratified queries at $\pm 20\%$ (the complete-interval companion of Figure 2).